Université Paris Dauphine
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Discrete processes

Teachers

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Evaluation

There are a mid-term exam (P) and a terminal exam (E). The final mark is given by $\max\{0.4P + 0.6E, E\}$.

Outline

A first version of these notes have been written together with P. Cardaliaguet and are largely inspired by those of F. Simenhaus, J. Lehec and F. Huveneers.

Throughout this course, $(\Omega, \mathcal{F}, \mathbb{P})$ denotes a probability space. A discrete process is a sequence of random variables $(X_n)_{n\geq 0}$ defined on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

This course is made up of three parts:

1. Conditional Expectation
2. Martingales
3. Markov chains

Please feel free to point out any errors in these notes!

Contents

1 Conditional expectation

1.1 Definition of conditional expectation

The notion of conditional expectation is given by the following theorem, which indicates that this notion is well defined. We fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G}$ a sub-σ-algebra of $\mathcal{F}$.

Theorem 1. *Let $X \in L^1(\mathcal{F})$. There is a unique random variable, denoted $\mathbb{E}(X|\mathcal{G})$, with the following two properties:*

1. *$\mathbb{E}(X|\mathcal{G})$ is measurable with respect to $\mathcal{G}$,*
2. *$\mathbb{E}(X|\mathcal{G})$ is integrable,*
3. *For any random variable Z bounded and measurable with respect to $\mathcal{G}$,*

$$\mathbb{E}(XZ) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})Z).$$

Remark 1. *Note that $\mathbb{E}(X|\mathcal{G})$ is a random variable, not a number. In particular, $\mathbb{E}(X|\mathcal{G})$ is only almost surely defined.*

Remark 2. *When $\mathcal{G} = \sigma(Y)$, where Y is a r.v., we write*

$$\boxed{\mathbb{E}(X|Y) \ := \ \mathbb{E}(X|\sigma(Y)).}$$

Note that, as any r.v. which is $\sigma(Y)-$measurable is a Borel function of Y (see Lemma 1 below), there exists a Borel function $f : \mathbb{R} \to \mathbb{R}$ such that $\mathbb{E}(X|Y) = f(Y)$.

Remark 3. *We will use the equivalence between the following conditions:*

1. *For any random variable Z bounded and measurable with respect to $\mathcal{G}$,*

$$\mathbb{E}(XZ) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})Z).$$

2. *For any event $A \in \mathcal{G}$,*

$$\mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})\mathbf{1}_A).$$

3. *For any random variable Z measurable with respect to $\mathcal{G}$ and such that XZ is integrable,*

$$\mathbb{E}(XZ) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})Z).$$

The second condition is often the easiest to verify. The last one is very useful.

Remark 4. *As in the classic framework of the Lebesgue integral, we can also define the conditional expectation for a nonnegative (but not necessarily integrable) r.v. X. We can then show the existence of a unique r.v., still denoted $\mathbb{E}(X|\mathcal{G})$, taking values in $[0, +\infty]$, such that*

1. $\mathbb{E}(X|\mathcal{G})$ *is* $\mathcal{G}-$*measurable,*

2. For any random variable Z *nonnegative and measurable with respect to* $\mathcal{G}$*,*

$$\mathbb{E}(XZ) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})Z)$$

(these two expectations can be equal to $+\infty$*)*

Theorem 1 will be proven later, after the discussion of some particular cases where the calculation of the conditional expectation is simple.

1.2 Discrete random variables

Conditioning with respect to an event: Recall that, if A and B are two events (i.e., $A, B \in \mathcal{F}$), with $\mathbb{P}(B)$ >0, then

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This naturally leads to defining the conditional expectation of an r.v. $X \in L^1(\mathcal{F})$ given B (with $B \in \mathcal{F}$ and $\mathbb{P}(B) > 0$) as

$$\mathbb{E}(X|B) = \frac{\mathbb{E}(X\mathbf{1}_B)}{\mathbb{P}(B)},$$

which corresponds to the expectation of X under the probability measure $\mathbb{P}(\cdot|B)$. Note the difference in nature with the conditional expectation with respect to a σ-algebra defined in Theorem 1: $\mathbb{E}(X|B)$ is a number while $\mathbb{E}(X|\mathcal{G})$ (where $\mathcal{G}$ is a σ-field) is a r.v..

The objective of this part is to explain the link between the two notions when $\mathcal{G} = \sigma(Y)$, with Y a discrete r.v.. A random variable Y on Ω is said to be *discrete* if it takes only a finite or countable number of values a.s., *i.e.*, $E = \{y \in \mathbb{R}; \mathbb{P}(Y = y) > 0\}$ is finite or countable and $\sum_{y \in E} \mathbb{P}(Y = y) = 1$. In other words, it staisfies

$$Y = \sum_{y \in E} y\mathbf{1}_{Y=y} \quad \text{a.s.}$$

Remark 5. *We denote by* $\mathbf{1}_A$ *the indicator function of a set* A*. We used the notation* $\mathbf{1}_{x=y}$ *to denote* $\mathbf{1}_{\{y\}}(x)$*.*

For instance, if Y follows a Bernoulli distribution with parameter $p \in (0, 1)$,

$$Y = 0.\mathbf{1}_{Y=0} + 1.\mathbf{1}_{Y=1}, \qquad \text{with } \mathbb{P}(Y = 0) = 1 - p > 0,\ \mathbb{P}(Y = 1) = p > 0.$$

If Y follows a Poisson distribution with parameter $\lambda > 0$, then

$$Y = \sum_{n=0}^{\infty} n\mathbf{1}_{Y=n}, \qquad \text{with } \mathbb{P}(Y = n) = \frac{\lambda^n}{n!}e^{-\lambda} > 0.$$

Let Y be a discrete r.v. and X an integrable r.v. . We will see that, in this case, the expression of $\mathbb{E}(X|Y)$ is explicit.

Proposition 1. *Let Y be a discrete r.v. and $X \in L^1(\mathcal{F})$. Consider the function*

$$u : \mathbb{R} \mapsto \mathbb{R}, y \mapsto u(y) = \begin{cases} \mathbb{E}(X|Y = y) & \text{if } \mathbb{P}(Y = y) > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}(X|Y) = u(Y),$$

i.e., explicitly

$$\mathbb{E}(X|Y) \ = \ \sum_{y \in E} \frac{\mathbb{E}(X \, \mathbf{1}_{Y=y})}{\mathbb{P}(Y = y)} \mathbf{1}_{Y=y}.$$

Remark 6. *The value of u on $N = \{y \in \mathbb{R};\ \mathbb{P}(Y = y) = 0\}$ is irrelevant as $\mathbb{P}(Y \in N) = 0$.*

Remark 7. *If X is also discrete, then we have another meaningful formulation for conditional expectation by observing that*

$$\mathbb{E}(X|Y = y) = \frac{\mathbb{E}(X \, \mathbf{1}_{Y=y})}{\mathbb{P}(Y = y)} = \sum_x x \, \mathbb{P}\left(X = x|Y = y\right),$$

which follows by writing $X = \sum_x x \mathbf{1}_{X=x}$ and permuting $\sum_x$ and $\mathbb{E}$.

To show the proposition, we will need to give an explicit form of the functions measurable with respect to $\sigma(Y)$. For this, we recall a very useful result from measure theory:

Lemma 1. *A random variable Z is measurable with respect to $\sigma(Y)$ if and only if there exists a Borelian function $f : \mathbb{R} \to \mathbb{R}$ such that $Z = f(Y)$.*

We come to the proof of the main result of this part.

Proof of Proposition 1. As u is a Borel function, $u(Y)$ is measurable with respect to $\sigma(Y)$. Moreover $u(Y)$ is integrable since

$$\mathbb{E}(|u(Y)|) = \sum_{y \in E} \frac{|\mathbb{E}(X \, \mathbf{1}_{Y=y})|}{\mathbb{P}(Y = y)} \mathbb{E}(\mathbf{1}_{Y=y}) \leq \sum_{y \in E} \mathbb{E}(|X| \, \mathbf{1}_{Y=y}) = \mathbb{E}(|X|) < \infty.$$

Finally let Z be a bounded r.v. which is measurable with respect to $\sigma(Y)$. According to Lemma 1, there exists a bounded Borel function $f : \mathbb{R} \to \mathbb{R}$ such that $Z = f(Y)$. So $Z = \sum_{y \in E} f(y) \mathbf{1}_{Y=y}$. Then (using by dominated convergence to permute sum and expectation if E is countable)

$$\mathbb{E}(XZ) = \sum_{y \in E} f(y) \mathbb{E}(X \mathbf{1}_{Y=y}) = \sum_{y \in E} f(y) \mathbb{E}(X|Y = y) \mathbb{P}(Y = y),$$

while

$$\begin{aligned} \mathbb{E}(u(Y)Z) = \sum_{y \in E} f(y) \mathbb{E}(u(Y) \mathbf{1}_{Y=y}) &= \sum_{y \in E} f(y) u(y) \mathbb{P}(Y = y) \\ &= \sum_{y \in E} f(y) \mathbb{E}(X|Y = y) \mathbb{P}(Y = y). \end{aligned}$$

Note that we can use the dominated convergence to permute sum and expectation if E is countable. As $\mathbb{E}(XZ) = \mathbb{E}(u(Y)Z)$ for any r.v. Z which is bounded and measurable with respect to $\sigma(Y)$, we conclude that $u(Y) = \mathbb{E}(X|Y)$ by the definition of conditional expectation. $\square$

1.3 Random Variables with a density

Another case where the calculation of the conditional expectation $\mathbb{E}(X|Y)$ is particularly simple is the case where the pair (X, Y) has a density. Consider a pair of variables (X, Y) with joint density $p_{X,Y}(x, y)$, and marginal densities $p_X(x)$ and $p_Y(y)$. We assume that $X \in L^1$. For all $y \in \mathbb{R}$ such that $p_Y(y) > 0$, we define the conditional density of X given $Y = y$ by

$$p_{X|Y=y}(y) \;=\; \frac{p_{X,Y}(x, y)}{p_Y(y)},$$

as well as the conditional expectation of X given $Y = y$ by

$$\mathbb{E}(X|Y = y) \;=\; \int_{\mathbb{R}} x p_{X|Y=y}(x)\mathrm{d}x.$$

Let us then define the real function u defined on the set $\{y \in \mathbb{R} : p_Y(y) > 0\}$ by $u(y) = \mathbb{E}(X|Y = y)$.

Proposition 2. *It holds*

$$\mathbb{E}(X|Y) \;=\; u(Y).$$

where

$$u : y \in \mathbb{R} \mapsto \begin{cases} \mathbb{E}(X|Y = y) & \text{if } p_Y(y) > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Remark 8. *The value of u on $N = \{y \in \mathbb{R};\ p_Y(y) = 0\}$ is irrelevant as $\mathbb{P}(Y \in N) = 0$.*

Proof. We need to check that $u(Y)$ satisfies the three points of Theorem/Definition 1. Assume first that u is a Borel map and let us check the integrability of $u(Y)$. Indeed, using Fubini–Tonelli Theorem, we have

$$\begin{aligned}\mathbb{E}(|u(Y)|) = \int_{\mathbb{R}} |u(y)| p_Y(y)\mathrm{d}x &\leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \frac{p_{X,Y}(x, y)}{p_Y(y)} |x| \mathrm{d}x \right) p_Y(y)\mathrm{d}y \\ &= \int_{\mathbb{R}^2} |x| p_{X,Y}(x, y)\mathrm{d}x\mathrm{d}y = \mathbb{E}[|X|] < +\infty.\end{aligned}$$

In addition, the above computation also induces that

$$\int_{\mathbb{R}} \left(\int_{\mathbb{R}} \frac{p_{X,Y}(x, y)}{p_Y(y)} |x| \mathrm{d}x \right) p_Y(y)\mathrm{d}y < +\infty,$$

and thus $u : \mathbb{R} \to \mathbb{R}$ is indeed a Borel map according to Fubini Theorem. We deduce that $u(Y)$ is measurable with respect to $\sigma(Y)$, according to Lemma 1.

Then let Z be a bounded random variable which is measurable with respect to $\sigma(Y)$. Still according to Lemma 1, there exists a Borelian function f such that $Z = f(Y)$. We compute

$$\begin{aligned}
\mathbb{E}(XZ) &= \mathbb{E}(Xf(Y)) \\
&= \int_{\mathbb{R}^2} xf(y)p_{X,Y}(x,y)\mathrm{d}x\mathrm{d}y \\
&= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} x \frac{p_{X,Y}(x,y)}{p_Y(y)} \mathrm{d}x \right) f(y)p_Y(y)\mathrm{d}y \\
&= \int_{\mathbb{R}} u(y)f(y)p_Y(y)\mathrm{d}y \\
&= \mathbb{E}(u(Y)f(Y)) \;=\; \mathbb{E}(u(Y)Z).
\end{aligned}$$

In this computation, we used the equality

$$p_{X,Y}(x,y) \;=\; \frac{p_{X,Y}(x,y)}{p_Y(y)} p_Y(y)$$

valid with the convention $p_{X,Y}(x,y)/p_Y(y) = 0$ if $p_Y(y) = 0$. $\square$

1.4 Random variables in $L^2(\Omega)$

We now start the proof of Theorem 1 on the existence and uniqueness of the conditional expectation. Let $\mathcal{G} \subset \mathcal{F}$ be a sub-σ-algebra of $\mathcal{F}$. We start by assuming that $X \in L^2(\mathcal{F})$. The following result says that $\mathbb{E}(X|\mathcal{G})$ is given by the orthogonal projection of X onto the space $L^2(\mathcal{G})$ of r.v. which are in $L^2(\mathcal{F})$ and which are measurable with respect to $\mathcal{G}$. Recall that $L^2(\mathcal{G})$ is a closed linear subspace of $L^2(\mathcal{F})$

Let us recall the projection theorem.

Theorem 2. *If $X \in L^2(\mathcal{F})$, then X uniquely decomposes into*

$$X \;=\; X^{\parallel} + X^{\perp}$$

where $X^{\parallel} \in L^2(\mathcal{G})$ and $X^{\perp} \in (L^2(\mathcal{G}))^{\perp}$ is orthogonal to any element of $L^2(\mathcal{G})$.

Proposition 3. *If $X \in L^2(\mathcal{F})$, then*

$$\mathbb{E}(X|\mathcal{G}) \;=\; X^{\parallel}.$$

Proof. Since $X^{\parallel} \in L^2(\mathcal{G})$ with $L^2(\mathcal{G}) \subset L^1(\mathcal{F})$, $X^{\parallel}$ is integrable and $\mathcal{G}-$measurable. Let us now show that, for any random variable $Z \in L^2(\mathcal{G})$,

$$\mathbb{E}(XZ) \;=\; \mathbb{E}(X^{\parallel}Z).$$

As $X = X^{\perp} + X^{\parallel}$ (with the notations of the projection theorem), we have, for any random variable Z $\mathcal{G}-$measurable and bounded (and therefore in $L^2(\mathcal{G})$),

$$\mathbb{E}(XZ) \ = \ \mathbb{E}(X^{\perp}Z) + \mathbb{E}(X^{\parallel}Z) \ = \ \mathbb{E}(X^{\parallel}Z).$$

$\square$

1.5 Random variables in $L^1(\Omega)$

We now proceed to the proof of the existence and uniqueness of the conditional expectation $\mathbb{E}(X|\mathcal{G})$ when Y is only integrable (Theorem 1). We denote by $L^1(\mathcal{G})$ the subspace of $L^1(\mathcal{F})$ of functions which are measurable with respect to $\mathcal{G}$. We recall that, as $\mathbb{P}$ is a probability measure, $L^2(\mathcal{F}) \subset L^1(\mathcal{F})$, and that the notion of orthogonal projection is not defined in L^1.

To prove Theorem 1, we will first need a monotonicity result:

Lemma 2. *Let $X_1, X_2 \in L^1(\mathcal{F})$ such that $X_1 \leq X_2$ a.s., let Y_1 verify the three points of Theorem 1 for X_1, and let Y_2 verify the two points of Theorem 1 for X_2. Then $Y_1 \leq Y_2$ a.s.*

Proof. Consider the variable $Z = \mathbf{1}_{(Y_1 - Y_2) \geq 0}$ which is measurable with respect to $\mathcal{G}$. We have $Z \geq 0$ a.s. and therefore, by hypothesis,

$$0 \ \geq \ \mathbb{E}((X_1 - X_2)Z) \ = \ \mathbb{E}((Y_1 - Y_2)Z) \ \geq \ 0$$

since $(Y_1 - Y_2)Z = (Y_1 - Y_2)\mathbf{1}_{(Y_1 - Y_2) \geq 0} \geq 0$ a.s. Therefore $\mathbb{E}((Y_1 - Y_2)Z) = 0$ and thus $(Y_1 - Y_2)Z = 0$ a.s. since this variable is non-negative. This implies $Y_1 \leq Y_2$ a.s. $\square$

We also recall the monotone convergence theorem: if $(X_n)_{n \geq 1}$ is an non-decreasing sequence of nonnegative random variables on Ω, and if the random variable X is its a.s. limit (possibly being infinite on a set of non-zero measures), then $\mathbb{E}(X_n) \to \mathbb{E}(X)$ as $n \to \infty$ (all these expressions possibly being infinite).

Proof of Theorem 1.

Uniqueness. Let Y_1 and Y_2 be two variables that would verify the result. By monotonicity, we have $Y_1 \leq Y_2$ and $Y_2 \leq Y_1$ a.s., and therefore $Y_1 = Y_2$ a.s.

Existence. First suppose $X \geq 0$ a.s. We define the sequence $(X_n)_{n \geq 1}$ by

$$X_n = X \wedge n, \quad n \geq 1.$$

The variables X_n are bounded and therefore in L^2, and thus we can define $Y_n = \mathbb{E}(X_n|\mathcal{G})$ for all $n \geq 1$, as the orthogonal projection of X_n onto $L^2(\mathcal{G})$

Let us observe that

1. The sequence (X_n) is nondecreasing and therefore, by monotonicity, the sequence (Y_n) is too.

2. The variables X_n are nonnegative and therefore, by monotonicity, the variables Y_n too.

By the monotone convergence theorem, we conclude that Y_n tends to a variable Y and that $\mathbb{E}(Y_n) \to \mathbb{E}(Y)$ as $n \to \infty$. But

$$0 \le \mathbb{E}(Y_n) = \mathbb{E}(X_n) \le \mathbb{E}(X)$$

where the equality comes from the fact that $\mathbb{E}(X_n) = \mathbb{E}(X_n^{\|}) + \mathbb{E}(X_n^{\perp}) = \mathbb{E}(Y_n) + \langle X_n^{\perp}, 1\rangle$ and the dot product is zero because 1 is measurable with respect to $\mathcal{G}$. We conclude that $\mathbb{E}(Y) < +\infty$ and by monotonicity and positivity, that the convergence of Y_n to Y takes place in L^1.

Let us show that Y has the three desired properties. First, Y is measurable with respect to $\mathcal{G}$ by passing to the limit. Then, for any Z bounded and measurable with respect to $\mathcal{G}$, we have

$$\begin{aligned}\mathbb{E}(XZ) &= \mathbb{E}((X - X_n)Z) + \mathbb{E}(X_n Z) \;=\; \mathbb{E}((X - X_n)Z) + \mathbb{E}(Y_n Z) \\ &= \mathbb{E}((X - X_n)Z) + \mathbb{E}((Y_n - Y)Z) + \mathbb{E}(YZ),\end{aligned}$$

and the first two terms on the right-hand side tend to 0 by convergence in L^1 as $n \to \infty$. Therefore, $\mathbb{E}(XZ) = \mathbb{E}(YZ)$.

Let us finally consider the general case $X \in L^1$. We decompose

$$X \;=\; X_+ - X_-,$$

where $X_+ \ge 0$ and $X_- \ge 0$. Each of these terms is in L^1. We apply the first part of the proof, to obtain the corresponding variables Y_+ and Y_-, and we check that $Y_+ - Y_-$ has the two desired properties. □

1.6 Some properties

We give some properties of the conditional expectation. The proofs are left as an exercise. Let $X, Y \in L^1(\mathcal{F})$ and let $\mathcal{G}$ be a sub-σ-algebra of $\mathcal{F}$. The first properties are common to (unconditional) expectation:

1. If $\lambda \in \mathbb{R}$, $\mathbb{E}(X + \lambda Y|\mathcal{G}) = \mathbb{E}(X|\mathcal{G}) + \lambda\mathbb{E}(Y|\mathcal{G})$ (linearity).
2. If $X \ge 0$ a.s., then $\mathbb{E}(X|\mathcal{G}) \ge 0$ a.s. (positivity).
3. (conditional Jensen) If $\varphi : \mathbb{R} \to \mathbb{R}$ is convex and measurable, and if $\varphi(X) \in L^1$, then $\varphi(\mathbb{E}(X|\mathcal{G})) \le \mathbb{E}(\varphi(X)|\mathcal{G})$ a.s. .

   (in particular, if $X \in L^p$, then $\mathbb{E}(X|\mathcal{G}) \in L^p$).
4. $\mathbb{E}(1|\mathcal{G}) = 1$.

The following five properties are specific to the conditional expectation:

1. If X is measurable with respect to $\mathcal{G}$, then $\mathbb{E}(X|\mathcal{G}) = X$.
2. If X is measurable with respect to $\mathcal{G}$ and if XY and Y are integrable, then $\mathbb{E}(XY|\mathcal{G}) = X\mathbb{E}(Y|\mathcal{G})$.
3. If X is independent of $\mathcal{G}$, then $\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$.
4. $\mathbb{E}(\mathbb{E}(X|\mathcal{G})) = \mathbb{E}(X)$.
5. If $\mathcal{G}' \subset \mathcal{G}$, then $\mathbb{E}(\mathbb{E}(X|\mathcal{G}')|\mathcal{G}) = \mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{G}') = \mathbb{E}(X|\mathcal{G}')$ (tower property).

The convergence properties of the conditional expectation are very similar to those of the classical expectation.

1. (convergence in L^p) $\mathbb{E}(X_n|\mathcal{G}) \to \mathbb{E}(X|\mathcal{G})$ in L^p if $X_n \to X$ in L^p (where $p \in [1, +\infty]$).
2. (monotone convergence) $\mathbb{E}(X_n|\mathcal{G}) \nearrow \mathbb{E}(X|\mathcal{G})$ if $0 \leq X_n \nearrow X$.
3. (Fatou) $\mathbb{E}(\liminf_n X_n|\mathcal{G}) \leq \liminf_n E(X_n|\mathcal{G})$ if $X_n \geq 0$.
4. (Dominated convergence) $\mathbb{E}(X_n|\mathcal{G}) \to \mathbb{E}(X|\mathcal{G})$ a.s. and in L^1 if $X_n \to X$ a.s. and $|X_n| \leq Z \in L^1$.

Finally, let us discuss a case where the computation of the conditional expectation boils down to the computation of an expectation:

Theorem 3. *We assume that X, Y are two r.v., with Y independent of $\mathcal{G}$ and X measurable with respect to $\mathcal{G}$. Let $\varphi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be bounded Borel-measurable. Then*

$$\mathbb{E}(\varphi(X, Y)|\mathcal{G}) = \Phi(X) \qquad \text{where } \Phi(x) = \mathbb{E}(\varphi(x, Y)) \; \forall x \in \mathbb{R}.$$

Proof. Note first that $\Phi(X)$ is measurable with respect to $\mathcal{G}$ and bounded. Let Z be a r.v. $\mathcal{G}-$measurable and bounded. The distribution $\mathbb{P}_{X,Y,Z}$ can be written, since Y is independent of (X, Z), as $\mathbb{P}_{X,Y,Z} = \mathbb{P}_Y \otimes \mathbb{P}_{X,Z}$. So, by Fubini,

$$\begin{aligned}\mathbb{E}(\varphi(X, Y)Z) = \int_{\mathbb{R}^3} \varphi(x, y) z d\mathbb{P}_{X,Y,Z}(x, y, z) &= \int_{\mathbb{R}^2} (\int_{\mathbb{R}} \varphi(x, y) d\mathbb{P}_Y(y)) z d\mathbb{P}_{X,Z}(x, z) \\ = \int_{\mathbb{R}^2} \mathbb{E}(\varphi(x, Y)) z d\mathbb{P}_{X,Z}(x, z) &= \int_{\mathbb{R}^2} \Phi(x) z d\mathbb{P}_{X,Z}(x, z) = \mathbb{E}(\Phi(X)Z).\end{aligned}$$

So $\mathbb{E}(\varphi(X, Y)|\mathcal{G}) = \Phi(X)$. □

1.7 Gaussian random variables

If $(X_0, X_1, \ldots, X_n)$ is a Gaussian vector, then $\mathbb{E}(X_0|\sigma(X_1, \ldots, X_n))$ is a linear combination of $X_1, \ldots, X_n$:

Proposition 4. *We assume that* $(X_0, X_1, \dots, X_n)$ *is a Gaussian vector. Then there exist real numbers* $c_0, c_1 \dots, c_n$ *such that*

$$\mathbb{E}(X_0|\sigma(X_1, \dots, X_n)) = c_0 + \sum_{i=1}^{n} c_i X_i.$$

This result is left as an exercise.

2 Martingales

2.1 Definitions et examples

First definitions. A sequence $(X_n)_{n\geq 0}$ of random variables is called a *stochastic process.* A sequence $(\mathcal{F}_n)_{n\geq 0}$ of σ-algebras is called a *filtration* if it is non-decreasing with respect to the inclusion:

$$\mathcal{F}_n \subset \mathcal{F}_{n+1} \qquad \forall n \geq 0.$$

We say that the process (X_n) is *adapted* to the filtration $(\mathcal{F}_n)$ if X_n is measurable with respect to $\mathcal{F}_n$ for all $n \geq 0$.

Interpretation. Typically, the index n corresponds to the time and $\mathcal{F}_n$ to the information available at time n. The property of filtration means that the information accumulates over time. A process is adapted when its value X_n is known at time n.

Example. Given a process $(X_n)_{n\geq 0}$, the sequence of σ-algebras $(\mathcal{F}_n)_{n\geq 0}$ defined by

$$\mathcal{F}_n = \sigma(X_0, \ldots, X_n),$$

is a filtration, where we recall that $\sigma(X_0, \ldots, X_n)$ is the smallest σ-algebra making the variables $X_0, \ldots, X_n$ measurable. It is called the *natural filtration* of the process (X_n). Observe that a process is always adapted to its natural filtration.

Definition of martingale. Let $(X_n)_{n\geq 0}$ be a process and $(\mathcal{F}_n)_{n\geq 0}$ a filtration. We say that (X_n) is a *martingale* with respect to $(\mathcal{F}_n)$ if, for all $n \geq 0$,

1. X_n is mesurable with respect to $\mathcal{F}_n$,
2. X_n is integrable,
3. $\mathbb{E}(X_{n+1}|\mathcal{F}_n) = X_n$.

We say that (X_n) is a *submartingale* (resp. *supermartingale*) if condition 3 above is replaced by $\mathbb{E}(X_{n+1}|\mathcal{F}_n) \geq X_n$ (resp. $\mathbb{E}(X_{n+1}|\mathcal{F}_n) \leq X_n$).

The first condition also reads: (X_n) is adapted to $(\mathcal{F}_n)$.

Interpretation. A martingale describes the evolution of a player's total gain in a fair game: At each time n, a round of the game occurs and brings the player a marginal gain $\Delta X_n = X_n - X_{n-1}$, the game is fair in the sense that $\mathbb{E}(\Delta X_n|\mathcal{F}_n) = 0$.

Remark 9. *We simply say that (X_n) is a martingale if (X_n) is a martingale with respect to a certain filtration. In this case, (X_n) is always a martingale with respect to its natural filtration (see Exercises).*

Let us give two elementary properties which follow directly from the definition.

Proposition 5. *Let* $(X_n)_{n\geq 0}$ *be a martingale with respect to a filtration* $(\mathcal{F}_n)_{n\geq 0}$. *Then it holds for all* $n \geq 0$,

1. $\mathbb{E}(X_n|\mathcal{F}_k) = X_k$ *for* $0 \leq k \leq n$,
2. $\mathbb{E}(X_n) = \mathbb{E}(X_0)$ *for* $n \geq 0$.

Proof. For the first part, we set $k \geq 0$ and we proceed by induction on $n \geq k$. The result is true for $n = k$ since X_k is measurable with respect to $\mathcal{F}_k$. Let us then assume it to be true for $n \geq k$ and prove it for $n+1$: by the tower property,

$$\mathbb{E}(X_{n+1}|\mathcal{F}_k) \ = \ \mathbb{E}(\mathbb{E}(X_{n+1}|\mathcal{F}_n)|\mathcal{F}_k) \ = \ \mathbb{E}(X_n|\mathcal{F}_k) \ = \ X_k.$$

For the second part, we proceed as follows:

$$\mathbb{E}(X_n) \ = \ \mathbb{E}(\mathbb{E}(X_n|\mathcal{F}_0)) \ = \ \mathbb{E}(X_0),$$

where the last equality follows from the first part. □

Proposition 6. *If* $(X_n)_{n\geq 0}$ *is a martingale and* $\varphi : \mathbb{R} \to \mathbb{R}$ *is a convex function such that* $\varphi(X_n) \in L^1$ *for all* $n \geq 0$, *then* $(\varphi(X_n))_{n\geq 0}$ *is a submartingale.*

Proof. It is an easy consequence of Jensen's inequality. Indeed it holds

$$\mathbb{E}(\varphi(X_{n+1})|\mathcal{F}_n) \geq \varphi(\mathbb{E}(X_{n+1}|\mathcal{F}_n)) = \varphi(X_n).$$

□

Example: the random walk on $\mathbb{Z}$. Let $(B_n)_{n\geq 1}$ be a sequence of i.i.d. random variable such that $\mathbb{P}(B_n = 1) = \mathbb{P}(B_n = -1) = \frac{1}{2}$. The process $(X_n)_{n\geq 0}$ defined by

$$\begin{cases} X_0 \ = \ 0, \\ X_{n+1} \ = \ X_n + B_{n+1}, \quad n \geq 0, \end{cases}$$

is a martingale. Note that

$$X_n \ = \ \sum_{k=1}^{n} B_k, \quad n \geq 1.$$

This example generalizes as follows: Given $(\Delta_n)_{n\geq 1}$ is a sequence of integrable i.i.d. random variables and $x \in \mathbb{R}$, the process $(X_n)_{n\geq 0}$ defined by $X_0 = x$ and

$$X_n \ = \ x + \sum_{k=1}^{n} \Delta_k - n\mathbb{E}(\Delta_1), \quad n \geq 1,$$

is a martingale (sometimes called a "generalized random walk") .

Example: multiplicative martingale. Let $(Y_n)_{n\geq 1}$ be a sequence of i.i.d. integrable random variables such that $\mathbb{E}(Y_n) = 1$ for all $n \geq 1$. Let also $x \in \mathbb{R}$. The process $(X_n)_{n\geq 0}$ defined by

$$\begin{cases} X_0 \;=\; x, \\ X_{n+1} \;=\; X_n Y_{n+1}, \quad n \geq 0, \end{cases}$$

is a martingale. Note that

$$X_n \;=\; x \prod_{k=1}^{n} Y_k, \quad n \geq 1.$$

Example: closed martingale. Let Z be an integrable random variable and $(\mathcal{F}_n)_{n\geq 0}$ a filtration. Then the process $(X_n)_{n\geq 0}$ defined by

$$X_n = \mathbb{E}(Z|\mathcal{F}_n), \quad n \geq 0,$$

is a martingale with respect to $(\mathcal{F}_n)$. Indeed, for all $n \geq 0$, $X_n \in L^1(\mathcal{F}_n)$ by definition of the conditional expectation and it follows from the tower property that

$$\mathbb{E}(X_{n+1}|\mathcal{F}_n) \;=\; \mathbb{E}(\mathbb{E}(Z|\mathcal{F}_{n+1})|\mathcal{F}_n) \;=\; \mathbb{E}(Z|\mathcal{F}_n) \;=\; X_n.$$

2.2 Stopping times

Definition of stopping time. We say that a random variable $T : \Omega \mapsto \mathbb{N} \cup \{+\infty\}$ is a *stopping time* with respect to a filtration $(\mathcal{F}_n)$ if

$$\{T = n\} \in \mathcal{F}_n, \qquad \text{for all } n \in \mathbb{N}.$$

We say that T is *finite* if $\mathbb{P}(T < +\infty) = 1$.

The condition $\{T = n\} \in \mathcal{F}_n$ for all $n \geq 0$ can be replaced by the condition

$$\{T \leq n\} \in \mathcal{F}_n \qquad \text{for all } n \geq 0.$$

Indeed, on the one hand, we have that

$$\{T \leq n\} = \{T = 0\} \cup \cdots \cup \{T = n\},$$

and $\{T = k\} \in \mathcal{F}_k \subset \mathcal{F}_n$ for all $0 \leq k \leq n$. On the other hand, it holds

$$\{T = n\} = \{T \leq n\} \cap \{T > n-1\} = \{T \leq n\} \cap \{T \leq n-1\}^{\mathsf{c}},$$

where $\{T \leq n\} \in \mathcal{F}_n$ and $\{T \leq n-1\}^{\mathsf{c}} \in \mathcal{F}_{n-1} \subset \mathcal{F}_n$.

Remark 10. *Do not confuse the notion of finite and bounded random variables: T is said to be bounded if there exists a number $M > 0$ such that $\mathbb{P}(T \leq M) = 1$. By definition, all random variables taking values in $\mathbb{N}$ or $\mathbb{R}_+$ are finite but they are not necessarily bounded For instance, a Poisson variable is not bounded but a binomial variable is.*

Interpretation. A stopping time corresponds to a legitimate stopping strategy for the game: The decision to quit the game or not should be based only upon the information available at time n. For instance,

- The player leaves as soon as his winnings are at least equal to 10:

$$T = \min\{n \geq 1 : X_n \geq 10\}.$$

The strategy is legitimate and T is a stopping time since

$$\{T \leq n\} = \{X_1 \geq 10\} \cup \cdots \cup \{X_n \geq 10\}, \quad n \geq 1.$$

- The player leaves the game just before losing for the first time:

$$T = \min\{n \geq 0 : \Delta X_{n+1} < 0\}.$$

The strategy is akin to cheating and T is not a stopping time.

Example: constant time. Let $k \in \mathbb{N}$, the constant time $T = k$ is a stopping time with respect to any filtration.

Example: hitting time of B. Let $B \subset \mathbb{R}$ be a Borel set and $(X_n)_{n\geq 0}$ a sequence of random variables. We set

$$T_B = \min\{n \geq 0 : X_n \in B\},$$

with the convention $\min \emptyset = +\infty$. The variable T_B is a stopping time with respect to the natural filtration of (X_n). Indeed, for all $n \geq 0$, we have

$$\{T_B \leq n\} = \{X_0 \in B\} \cup \cdots \cup \{X_n \in B\}$$

and $\{X_k \in B\} \in \mathcal{F}_k \subset \mathcal{F}_n$ for all $0 \leq k \leq n$.

2.3 Martingale transform and stopped process

Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration and $(X_n)_{n\geq 0}$ a martingale with respect to $(\mathcal{F}_n)$. Let us introduce a transformation of (X_n) which preserves the martingale property. For $n \geq 1$, we write

$$X_n = X_0 + \sum_{k=1}^{n} \Delta X_k, \quad \text{with } \Delta X_k = X_k - X_{k-1}, \quad k \geq 1.$$

We say that $(C_n)_{n\geq 1}$ is a *predictable* process when C_n is measurable with respect to $\mathcal{F}_{n-1}$ for all $n \geq 1$.

Proposition 7. *Let $(X_n)_{n\geq 0}$ be a martingale and $(C_n)_{n\geq 1}$ a predictable process such that $C_n \in L^\infty$ for all $n \geq 1$. Let also $Y_0 \in L^1(\mathcal{F}_0)$. Then the process $(Y_n)_{n\geq 0}$ defined by*

$$Y_n = Y_0 + \sum_{k=1}^{n} C_k \Delta X_k, \quad n \geq 1,$$

is a martingale.

Proof. It is easy to check that Y_n is integrable and measurable with respect to $\mathcal{F}_n$ for all $n \geq 0$. Moreover, it holds for all $n \geq 1$

$$\mathbb{E}(Y_n|\mathcal{F}_{n-1}) \;=\; \mathbb{E}(Y_{n-1} + C_n \Delta X_n|\mathcal{F}_{n-1}) \;=\; Y_{n-1} + C_n \mathbb{E}(\Delta X_n|\mathcal{F}_{n-1}),$$

since Y_{n-1} and C_n are measurable with respect to $\mathcal{F}_{n-1}$. In addition, we have

$$\mathbb{E}(\Delta X_n|\mathcal{F}_{n-1}) = \mathbb{E}(X_n|\mathcal{F}_{n-1}) - X_{n-1} = 0,$$

and so $\mathbb{E}(Y_n|\mathcal{F}_{n-1}) = Y_{n-1}$. □

Example: modified random walk. Let $y \in \mathbb{Z}$ and $(B_n)_{n\geq 1}$ a sequence of i.i.d. random variables such that $\mathbb{P}(B_n = -1) = \mathbb{P}(B_n = +1) = 1/2$. Let also $D : \mathbb{Z} \to \mathbb{N}$ be a bounded function. We consider the process $(Y_n)_{n\geq 0}$ defined by

$$\begin{cases} Y_0 \;=\; y, \\ Y_{n+1} \;=\; Y_n + D(Y_n) B_{n+1}. \end{cases}$$

For $n \geq 1$, we have

$$Y_n \;=\; y + \sum_{k=1}^{n} D(Y_{k-1}) B_k \;=\; y + \sum_{k=1}^{n} D(Y_{k-1}) \Delta X_k,$$

where (X_n) is a simple random walk. Therefore the process (Y_n) is a martingale.

Example: stopped process. Given a process $(X_n)_{n\geq 0}$ and a random time $T : \Omega \to \mathbb{N} \cup \{+\infty\}$, we define the *stopped process* $(X_{n\wedge T})_{n\geq 0}$ by

$$X_{n\wedge T} \;=\; \begin{cases} X_n & \text{if } n < T, \\ X_T & \text{otherwise.} \end{cases}$$

For example, when T is the hitting time of a Borel set B, $X_{n\wedge T}$ is equal to X_n until the process attains B, and then it remains constant, equal to the value X_T of the process when it attains B.

If (X_n) is a martingale and T is a stopping time with respect to the same filtration $(\mathcal{F}_n)$, then $(X_{n\wedge T})$ is still a martingale. Indeed, for $n \geq 1$, it holds

$$X_{n\wedge T} \;=\; X_0 + \sum_{k=1}^{n\wedge T} \Delta X_k \;=\; X_0 + \sum_{k=1}^{n} \mathbf{1}_{\{T\geq k\}} \Delta X_k, \quad n \geq 1,$$

and $\{T \geq k\} = \{T \leq k-1\}^c \in \mathcal{F}_{k-1}$ for all $k \geq 1$.

2.4 Optional stopping theorem

Let $(X_n)_{n\geq 0}$ be a martingale and T a finite stopping time with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$. It follows from the previous section that the stopped process $(X_{n\wedge T})$ is a martingale and so we have that

$$\mathbb{E}(X_{n\wedge T}) \;=\; \mathbb{E}(X_{0\wedge T}) \;=\; \mathbb{E}(X_0).$$

The question is whether we can pass to the limit to prove that

$$\mathbb{E}(X_T) \;=\; \mathbb{E}(X_0)\ ?$$

The answer is no in general. For instance, consider (X_n) a simple random walk and the stopping time

$$T \;=\; \min\{n \geq 0 : X_n = 1\}.$$

Then we can show that T is finite (see below for a proof). Nonetheless, since $X_T = 1$ by definition, it holds

$$1 \;=\; \mathbb{E}(X_T) \;\neq\; \mathbb{E}(X_0) = 0.$$

The celebrated *optional stopping theorem* provides conditions to guarantee that the equality holds:

Theorem 4 (Doob's optional stopping theorem)**.** *Let $(X_n)_{n\geq 0}$ be a martingale and T a stopping time with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$. Assume that one of the following conditions holds:*

1. *T is bounded*
2. *$\mathbb{E}(T) < +\infty$ and $(\Delta X_n)_{n\geq 1}$ is bounded in L^∞.*
3. *T is finite and $(X_{n\wedge T})_{n\geq 0}$ is bounded in L^∞.*

Then $X_T \in L^1$ and

$$\mathbb{E}(X_T) \;=\; \mathbb{E}(X_0).$$

Proof. Let us consider the three cases separately.

1. Suppose T is bounded, *i.e.*, there exists an integer N (deterministic) such that $T \leq N$ a.s.. In this case, it follows from the fact that the stopped process $(X_{n\wedge T})$ is a martingale that

$$\mathbb{E}(X_T) \;=\; \mathbb{E}(X_{N\wedge T}) \;=\; \mathbb{E}(X_{0\wedge T}) \;=\; \mathbb{E}(X_0).$$

2. The proof relies on the dominated convergence theorem. First we observe that, since T is integrable, T is finite and so $X_{n\wedge T}$ converges a.s. to X_T. Then, by assumption, there exists a number $M > 0$ such that $|\Delta X_k| \leq M$ a.s. for all $k \geq 1$ and therefore

$$|X_{n\wedge T}| \;=\; \left|X_0 + \sum_{k=1}^{n\wedge T} \Delta X_k\right| \;\leq\; |X_0| + M(n\wedge T) \;\leq\; |X_0| + MT$$

By assumption, $\mathbb{E}(|X_0| + MT) < +\infty$ and therefore, by dominated convergence, $X_T \in L^1$ and

$$\mathbb{E}(X_T) \;=\; \lim_{n\to\infty} \mathbb{E}(X_{n\wedge T}) \;=\; \mathbb{E}(X_0).$$

3. We proceed in a similar way, this time using the domination $|X_{n\wedge T}| \leq M$ a.s., valid for a certain number $M > 0$ and for all $n \geq 0$. $\square$

Remark 11. *The list of conditions given in Theorem 4 is by no means exhaustive: If T is finite and we can find $Z \in L^1$ such that $|X_{n\wedge T}| \leq Z$, then the dominated convergence theorem ensures that $\mathbb{E}[X_T] = \mathbb{E}[X_0]$.*

Example: random walk. This theorem allows us to derive several interesting facts about the random walk $(X_n)_{n\geq 0}$. Let $a \leq -1$ and $b \geq 1$ be two integers and define the stopping time

$$T \ = \ \min\{n \geq 0 : X_n = a \text{ or } X_n = b\},$$

which represents the exit time from the interval $]a, b[$.

Let us start by showing that T is finite. Let $\ell = b - a - 1$ which represents the maximum distance that the random walk must travel to exit $]a, b[$ if it has not yet exited. For all $n \in \mathbb{N}$, we have

$$\mathbb{P}(|X_{n+\ell} - X_n| = \ell) \ = \ 2\left(\frac{1}{2}\right)^{\ell} =: p > 0.$$

We observe that, if $T > m\ell$ for a certain integer $m \geq 1$, then $|X_{j+\ell} - X_j| < \ell$ for all $0 \leq j \leq (m-1)\ell$. So in particular, for all $m \geq 1$,

$$\mathbb{P}(T > m\ell) \ \leq \ \mathbb{P}\left(|X_{(k+1)\ell} - X_{k\ell}| < \ell, 0 \leq k \leq m-1\right).$$

The right-hand side corresponds to the probability of an intersection of m independent events, each of probability $1 - p$, and therefore

$$\mathbb{P}(T > m\ell) \ \leq \ (1-p)^m,$$

which implies that T is finite since

$$\mathbb{P}(T = +\infty) = \lim_{m\to\infty} \mathbb{P}(T > m\ell) = 0.$$

Let us note that this computation also implies that $\mathbb{E}(T) < +\infty$ by using the formula $\mathbb{E}(T) = \sum_{n\geq 0} \mathbb{P}(T > n)$.

We can now compute the probability that the random walk leaves this interval at a or at b, *i.e.*, $\mathbb{P}(X_T = a)$ and $\mathbb{P}(X_T = b)$. As the process $(X_{n\wedge T})_{n\geq 0}$ is bounded in L^∞, we can apply the optional stopping theorem and conclude that $\mathbb{E}(X_T) = \mathbb{E}(X_0) = 0$. We obtain

$$\begin{cases} \mathbb{P}(X_T = a)a + \mathbb{P}(X_T = b)b \ = \ 0 \\ \mathbb{P}(X_T = a) + \mathbb{P}(X_T = b) \ = \ 1 \end{cases}$$

and thus

$$\mathbb{P}(X_T = a) \ = \ \frac{b}{b-a}, \qquad \mathbb{P}(X_T = b) \ = \ \frac{-a}{b-a}.$$

We can also compute $\mathbb{E}(T)$. We start by noticing that the process $(Y_n)_{n\geq 0}$, defined by $Y_n = X_n^2 - n$ for $n \geq 0$, is a martingale. Even if none of the three assumptions of the optional stopping theorem apply, we can still prove that $\mathbb{E}(Y_T) = \mathbb{E}(Y_0) = 0$. Indeed, we observe that

$$|Y_{n\wedge T}| \;\leq\; (\max(|a|, b))^2 + n \wedge T \;\leq\; (\max(|a|, b))^2 + T.$$

Since $\mathbb{E}(T) < +\infty$ and $Y_{n\wedge T} \to Y_T$ a.s., we can apply the dominated convergence to obtain

$$0 = \mathbb{E}(Y_{n\wedge T}) \to \mathbb{E}(Y_T) = \mathbb{E}(X_T^2 - T).$$

We conclude that

$$\mathbb{E}(T) \;=\; \mathbb{P}(X_T = a)a^2 + \mathbb{P}(X_T = b)b^2 \;=\; |ab|.$$

Finally, let us consider the stopping time

$$T_1 \;=\; \min\{n \geq 0 : X_n = 1\},$$

and show that T_1 is finite as claimed at the beginning of this section. Taking $b = 1$ and $a \leq -1$ in the definition of T above, we have that

$$\mathbb{P}(T_1 < +\infty) \;\geq\; \mathbb{P}(X_T = 1) = \frac{-a}{1-a}, \qquad \text{for all } a \leq -1.$$

Therefore $\mathbb{P}(T_1 < +\infty) = 1$ by considering the limit $a \to -\infty$ in the inequality above. We can also conclude that $\mathbb{E}(T_1) = +\infty$ because otherwise the optional stopping theorem would imply that $1 = \mathbb{E}(X_{T_1}) = E(X_0) = 0$, which is a contradiction.

2.5 Maximal inequalities

Let (X_n) be a simple random walk on $\mathbb{Z}$. We can explicitly compute the distribution of X_n at a fixed time n, and conclude that X_n is "of order $\sqrt{n}$": for instance,

$$\mathbb{E}(X_n^2) = n \quad \text{and} \quad \frac{X_n}{\sqrt{n}} \xrightarrow[n\to+\infty]{\text{law}} \mathcal{N}(0,1).$$

Here we provide results which allow us to estimate $\max_{0\leq k\leq n} |X_k|$, the radius of the smallest interval centered at 0 in which the random walk can be enclosed until time n. We will see that $\max_{0\leq k\leq n} |X_k|$ is also "of order $\sqrt{n}$".

We recall that a process $(Y_n)_{n\geq 0}$ is a submartingale with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$ if $Y_n \in L^1(\mathcal{F}_n)$ and $\mathbb{E}(Y_{n+1}|\mathcal{F}_n) \geq Y_n$ for all $n \geq 0$. Let us start by an improvement of the Markov inequality for nonnegative submartingales:

Theorem 5. *Let $(Y_n)_{n\geq 0}$ be a nonnegative submartingale and $a > 0$. Then it holds for all $n \geq 0$,*

$$\mathbb{P}\left(\max_{0\leq k\leq n} Y_k \geq a\right) \;\leq\; \frac{\mathbb{E}(Y_n)}{a}.$$

In particular, this inequality holds for $Y_n = |X_n|$ where (X_n) is a martingale.

Proof. Denote by T the hitting time of $(a, +\infty)$, *i.e.*,

$$T = \min\{n \geq 0, X_n \geq a\}.$$

We start by using Markov inequality to obtain that

$$\mathbb{P}\left(\max_{0\leq k\leq n} Y_k \geq a\right) = \mathbb{P}(Y_{n\wedge T} \geq a) \leq \frac{\mathbb{E}(Y_{n\wedge T})}{a}.$$

It remains to show that $\mathbb{E}(Y_{n\wedge T}) \leq \mathbb{E}(Y_n)$ for all $n \geq 0$. This inequality is obvious for $n = 0$. Then we can compute for $n \geq 1$,

$$\begin{aligned}
\mathbb{E}(Y_n - Y_{n\wedge T}) &= \mathbb{E}((Y_n - Y_{n\wedge T})\mathbf{1}_{\{T<n\}}) \\
&= \sum_{k=0}^{n-1} \mathbb{E}((Y_n - Y_k)\mathbf{1}_{\{T=k\}}) \\
&= \sum_{k=0}^{n-1} \mathbb{E}(\mathbb{E}(Y_n - Y_k|\mathcal{F}_k)\mathbf{1}_{\{T=k\}}) \\
&= \sum_{k=0}^{n-1} \mathbb{E}((\mathbb{E}(Y_n|\mathcal{F}_k) - Y_k)\mathbf{1}_{\{T=k\}}) \geq 0,
\end{aligned}$$

since $\mathbb{E}(Y_n|\mathcal{F}_k) \geq Y_k$ for $0 \leq k \leq n-1$ as (Y_n) is a submartingale. □

Theorem 5 yields to a nice inequality on the moments of the maximum of a nonnegative submartingale.

Proposition 8. *Let $(Y_n)_{n\geq 0}$ be a nonnegative submartingale and $p > 1$. Then it holds for all $n \geq 0$,*

$$\mathbb{E}\left[\left(\max_{0\leq k\leq n} Y_k\right)^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}\left[Y_n^p\right].$$

In particular, this inequality holds for $Y_n = |X_n|$ where (X_n) is a martingale.

Proof. See Exercises. □

Example: the random walk. Let (X_n) be a simple random walk on $\mathbb{Z}$. Since the map $u \mapsto |u|$ is convex and nonnegative, the conditional Jensen inequality implies that $(|X_n|)$ is a nonnegative submartingale. Therefore, for all $n \geq 1$ and $K > 0$ (but the result is only interesting for $K > 1$), we obtain

$$\mathbb{P}\left(\max_{1\leq k\leq n} |X_k| \geq K\sqrt{n}\right) \leq \frac{\mathbb{E}(|X_n|)}{K\sqrt{n}} \leq \frac{\sqrt{\mathbb{E}(X_n^2)}}{K\sqrt{n}} = \frac{1}{K}.$$

Let us go further and improve substantially this inequality when K is large. We start by observing that for any $a > 0$,

$$\mathbb{P}\left(\max_{1\leq k\leq n} |X_k| \geq a\right) \leq \mathbb{P}\left(\max_{1\leq k\leq n} X_k \geq a\right) + \mathbb{P}\left(\min_{1\leq k\leq n} X_k \leq -a\right)$$

where, by symmetry of the random walk,

$$\mathbb{P}\left(\min_{1\le k\le n} X_k \le -a\right) = \mathbb{P}\left(\max_{1\le k\le n} (-X_k) \ge a\right) = \mathbb{P}\left(\max_{1\le k\le n} X_k \ge a\right).$$

Furthermore, since the map $u \mapsto \mathrm{e}^{\lambda u}$ is convex and nonnegative for any $\lambda > 0$, $(\mathrm{e}^{\lambda X_n})$ is also a nonnegative submartingale and it follows from the maximal inequality that

$$\begin{aligned}\mathbb{P}\left(\max_{1\le k\le n} |X_k| \ge a\right) \;&\le\; 2\mathbb{P}\left(\max_{1\le k\le n} X_k \ge a\right)\\ &=\; 2\mathbb{P}\left(\max_{1\le k\le n} \mathrm{e}^{\lambda X_k} \ge \mathrm{e}^{\lambda a}\right) \;\le\; 2\mathrm{e}^{-\lambda a}\mathbb{E}(\mathrm{e}^{\lambda X_n}).\end{aligned}$$

In addition, we have that

$$\mathrm{e}^{-\lambda a}\mathbb{E}(\mathrm{e}^{\lambda X_n}) \;=\; \mathrm{e}^{-\lambda a}\cosh(\lambda)^n \;\le\; \mathrm{e}^{\frac{\lambda^2}{2}n-\lambda a},$$

where we used the inequality $\cosh(x) \le \mathrm{e}^{x^2/2}$ for all $x \in \mathbb{R}$. This expression is minimal for $\lambda = a/n$ and is then equal to $\mathrm{e}^{-a^2/2n}$. Choosing $a = K\sqrt{n}$ for $K > 0$, we conclude that

$$\mathbb{P}\left(\max_{1\le k\le n} |X_k| \ge K\sqrt{n}\right) \;\le\; 2\mathrm{e}^{-K^2/2}.$$

2.6 Convergence of martingales

The goal is the section is to study the convergence of martingales as time increases to infinity. We start by discussing the convergence in L^p for $p > 1$.

Theorem 6. *If $(X_n)_{n\ge 0}$ is a martingale bounded in L^2, then X_n converges in L^2.*

Proof. The idea of the proof is to show that (X_n) is a Cauchy sequence in the Hilbert space L^2. First observe that for all $m > n > 0$,

$$X_n \;=\; X_0 + \sum_{k=1}^{n} \Delta X_k \quad \text{and} \quad X_m - X_n \;=\; \sum_{k=n+1}^{m} \Delta X_k.$$

In addition it holds for all $k \ge 1$,

$$\mathbb{E}(X_0 \Delta X_k) \;=\; \mathbb{E}(X_0 \mathbb{E}(\Delta X_k | \mathcal{F}_0)) \;=\; 0,$$

and, for all $l > k \ge 1$,

$$\mathbb{E}(\Delta X_k \Delta X_l) \;=\; \mathbb{E}(\Delta X_k \mathbb{E}(\Delta X_l | \mathcal{F}_k)) \;=\; 0.$$

Therefore, for all $m > n > 0$,

$$\mathbb{E}(X_n^2) = \mathbb{E}(X_0^2) + \sum_{k=1}^{n} \mathbb{E}(\Delta X_k^2) \quad \text{and} \quad \mathbb{E}((X_m - X_n)^2) = \sum_{k=n+1}^{m} \mathbb{E}(\Delta X_k^2).$$

Since $\sup_{n\in\mathbb{N}} \mathbb{E}(X_n^2) < +\infty$ by assumption, the first equality implies that the series $\sum_{k\geq 1} \mathbb{E}(\Delta X_k^2)$ converges. The conclusion then follows from the second equality which implies that (X_n) is Cauchy in L^2. $\square$

Remark 12. *More generally, we can show that, for any $p > 1$, a martingale bounded in L^p converges in L^p (see Exercises).*

Examples:

1. If $Z \in L^2$ and $(\mathcal{F}_n)$ is a filtration, then the closed martingale $(\mathbb{E}(Z|\mathcal{F}_n))_n$ is bounded in L^2 by $\mathbb{E}(Z^2)$, and therefore it converges in L^2.

2. If (X_n) is a simple random walk, then $X_n \in L^2$ and $E(X_n^2) = n$ for all $n \geq 0$, but it is not bounded in L^2, and so the theorem does not guarantee that (X_n) converges in L^2. This not surprising as we know that it does not even converge in law.

Let us turn now to the almost sure convergence of martingales. The next theorem provides a simple sufficient condition and is often used in the form of the following corollary. The proof is omitted and we refer to the lecture notes of F. Simenhaus for more details.

Theorem 7. *If $(X_n)_{n\geq 0}$ is a (sub/super)martingale bounded in L^1, then (X_n) converges almost surely to a variable $X \in L^1$.*

Corollary 1. *If $(X_n)_{n\geq 0}$ is a nonnegative supermartingale (resp. nonpositive submartinagle), then it converges almost surely to a variable $X \in L^1$.*

Proof. Let (X_n) be a nonnegative supermartingale. Then it is bounded in L^1 as

$$\mathbb{E}[|X_n|] = \mathbb{E}[X_n] \leq \mathbb{E}[X_0], \quad n \geq 1.$$

$\square$

Remark 13. *If (X_n) is bounded in L^2, it is also bounded in L^1. Therefore, by combining the two previous results, we conclude that a martingale bounded in L^2 converges in L^2 and almost surely.*

Remark 14. *A martingale bounded in L^1 does not necessarily converge in L^1. See examples below.*

Example: multiplicative martingale. Consider the multiplicative martingale given by $X_0 = 1$ and

$$X_n = Y_1 \cdots Y_n, \quad n \geq 1,$$

with (Y_n) i.i.d. such that $\mathbb{P}(Y_n = 2) = \mathbb{P}(Y_n = 0) = 1/2$. It is clearly nonnegative and, as $\mathbb{E}(X_n) = \mathbb{E}(X_0) = 1$ for all $n \geq 0$, it is also bounded in L^1. Thus (X_n) converges almost surely to a random variable $X \in L^1$. Actually we can identify $X = 0$ since

$$\mathbb{P}(X_n > 0) = \mathbb{P}(X_n = 2^n) = \frac{1}{2^n} \xrightarrow[n\to+\infty]{} 0.$$

However (X_n) does not converge to 0 in L^1 since $\mathbb{E}(|X_n - 0|) = \mathbb{E}(X_n) = 1$ for all $n \geq 0$.

Example: branching process. Let $(Y_{n,k})_{n\geq 0, k\geq 1}$ be a family of i.i.d. random variables valued in $\mathbb{N}$ such that

$$0 \;<\; \overline{Y} \;:=\; \mathbb{E}(Y_{0,1}) \;<\; +\infty.$$

We consider a process $(X_n)_{n\geq 0}$ defined by

$$\begin{cases} X_0 \;=\; 1, \\ X_{n+1} \;=\; \sum_{k=1}^{X_n} Y_{n,k}, \quad n \geq 0, \end{cases}$$

with the convention $\sum_{k=1}^{0}(\dots) = 0$. Typically X_n represents the size of a population at time n, where each individual k gives birth to a random number $Y_{n,k}$ of children forming the population at time $n+1$.

One can check that the sequence $(Z_n)_{n\geq 0}$, defined by $Z_n = X_n/\overline{Y}^n$ for $n \geq 0$, is a martingale, and so $\mathbb{E}(X_n) = \overline{Y}^n$ for all $n \geq 0$. Therefore the average size of the population decreases exponentially for $\overline{Y} < 1$, remains constant and equal to 1 for $\overline{Y} = 1$, and grows exponentially for $\overline{Y} > 1$.

In the case $\overline{Y} < 1$, we can easily show that (X_n) converges to 0 almost surely. Indeed, we have that for any $\varepsilon > 0$,

$$\mathbb{P}(X_n > \varepsilon) \;\leq\; \frac{\mathbb{E}(X_n)}{\varepsilon} \;=\; \frac{\overline{Y}^n}{\varepsilon},$$

and thus

$$\sum_{n=0}^{\infty} \mathbb{P}(X_n > \varepsilon) \;<\; +\infty,$$

which implies that $X_n \to 0$ a.s. by Borel–Cantelli Lemma.

In the case $\overline{Y} = 1$ and $\mathbb{P}(Y_{0,1} = 1) < 1$, we can still show that (X_n) converges almost surely to 0 but it is more involved. First notice that we necessarily have $\mathbb{P}(Y_{0,1} = 0) =: p > 0$. Next, we observe that (X_n) is a nonnegative martingale and, as $\mathbb{E}(X_n) = 1$ for all $n \geq 0$, it is also bounded in L^1. It follows that X_n converges almost surely to a random variable $X \in L^1$. Then it holds for $n \geq 0$,

$$\begin{aligned}
\mathbb{P}(X_{n+1} = 0) &= \sum_{k=0}^{\infty} \mathbb{P}(X_{n+1} = 0, X_n = k) = \sum_{k=0}^{\infty} \mathbb{P}(X_n = k) p^k \\
&= \mathbb{P}(X_n = 0) + \sum_{k=1}^{\infty} \mathbb{P}(X_n = k) p^k.
\end{aligned}$$

Since $\mathbb{P}(X_n = 0)$ converges when $n \to \infty$, we necessarily have $\mathbb{P}(X_n = k) \to 0$ for all $k \geq 1$. Therefore, $\mathbb{P}(X = k) = 0$ for all $k \geq 1$ and, as the support of X is in $\mathbb{N}$, we conclude that $X = 0$.

Remark 15. *The case $\overline{Y} = 1$ and $\mathbb{P}(Y_{0,1} = 1) < 1$ provides an additional example in which (X_n) converges almost surely to $X \in L^1$, but does not converge in L^1. It also gives a counterexample to the optional sampling theorem by considering the stopping time $T = \min\{n \geq 0 : X_n = 0\}$. Indeed, T is finite as $X_n \to 0$ a.s., but $0 = \mathbb{E}(X_T) \neq \mathbb{E}(X_0) = 1$.*

3 Markov Chains

3.1 Definitions and examples

Definitions. Let E be a countable space, that is to say finite or infinitely countable, which we call *space of states*. Let $(X_n)_{n\geq 0}$ be a process taking values in E. We say that (X_n) is a *Markov chain* if, for all $n \geq 0$, for all $y \in E$ and for all $x_0, \ldots, x_{n-1}, x \in E$ such that $\mathbb{P}(X_0 = x_0, \ldots, X_{n-1} = x_{n-1}, X_n = x) > 0$, we have

$$\mathbb{P}(X_{n+1} = y | X_0 = x_0, \ldots, X_{n-1} = x_{n-1}, X_n = x) \ = \ \mathbb{P}(X_{n+1} = y | X_n = x).$$

If, in addition, there exists a map $P : E \times E \to [0, 1]$ such that

$$\mathbb{P}(X_{n+1} = y | X_n = x) \ = \ P(x, y),$$

for all $n \geq 0$, and for all $x, y \in E$ such that $\mathbb{P}(X_n = x) > 0$, we say that (X_n) is *homogeneous in time*. In this case, P is called the *transition matrix* of the process (X_n). It is a so-called *stochastic matrix*, *i.e.*,

$$P(x, y) \ \geq \ 0 \quad \text{and} \quad \sum_{y \in E} P(x, y) \ = \ 1, \quad \forall x, y \in E.$$

In this course, we will only consider time-homogeneous Markov chains, and we will therefore no longer specify it.

Examples.

1. It may be convenient to make a graphical representation of the transition matrix of a Markov chain. For example, in the case where $E = \{1, 2, 3, 4\}$ and where the transition matrix is given by

$$P \ = \ \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1/3 & 0 & 2/3 \\ 3/4 & 0 & 0 & 1/4 \end{pmatrix},$$

   We obtain the graph in figure 1 below.

2. *Random walk in $\mathbb{Z}^d$.* Let $d \in \mathbb{N}^\star$ and $E = \mathbb{Z}^d$. Let $(B_n)_{n\geq 1}$ be a sequence of i.i.d. variables uniformly distributed on $\{\pm e_1, \ldots, \pm e_d\}$ where $(e_1, \ldots, e_d)$ represents the canonical basis of $\mathbb{R}^d$. We define the process $(X_n)_{n\geq 0}$ valued in $\mathbb{Z}^d$ by

$$\begin{cases} X_0 \ = \ 0 \\ X_{n+1} \ = \ X_n + B_{n+1}, \quad n \geq 0. \end{cases}$$

   This process is a Markov chain and its transition matrix is given by

$$P(x, y) \ = \ \frac{1}{2d} 1_{\{x \sim y\}}$$

   where $x \sim y$ if and only if $|x - y|_1 = 1$, that is, if x and y are close neighbors on $\mathbb{Z}^d$.

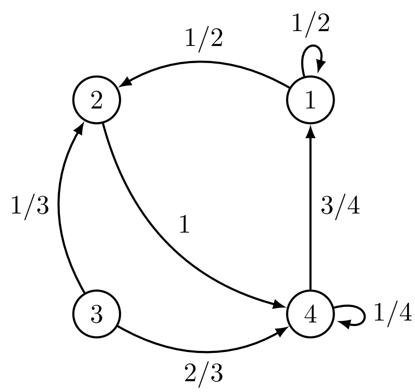

Figure 1: The graph representing the matrix P defined in the first example.

3. Let us modify the random walk in $\mathbb{Z}$ to make it a non-Markov process. Let $(B_n, B'_n)_{n\geq 1}$ be a sequence of i.i.d. pairs of random variables. We assume that $B_n, B'_n \in \{-1, 1\}$ a.s. and that

$$\mathbb{P}(B_n = 1) = p, \qquad \mathbb{P}(B'_n = 1) = p',$$

for two numbers $0 < p < p' < 1$. We define the process $(X_n)_{n\geq 0}$ as follows:

$$\begin{cases} X_0 \;=\; 0, \quad X_1 \;=\; B_1, \\ X_{n+1} \;=\; X_n + 1_{\{X_n > X_{n-1}\}} B_{n+1} + 1_{\{X_n < X_{n-1}\}} B'_{n+1}, \quad n \geq 1. \end{cases}$$

This process does not have the Markov property since

$$\begin{aligned} \mathbb{P}(X_3 = 1 | X_1 = 1, X_2 = 0) \;&=\; \mathbb{P}(B'_3 = 1) \;=\; p', \\ \mathbb{P}(X_3 = 1 | X_1 = -1, X_2 = 0) \;&=\; \mathbb{P}(B_3 = 1) \;=\; p. \end{aligned}$$

However, the law of X_{n+1} does not depend on the entire past of the process, but only on X_n and X_{n-1}. In a case like this, we can find a Markov process by introducing a second process (Y_n), defined here by

$$Y_0 \;=\; -1, \qquad Y_{n+1} = X_n, \quad n \geq 0.$$

The process (X_n, Y_n) is then a Markov process on $\mathbb{Z}^2$, since for all $n \geq 0$, we can write

$$\begin{cases} X_{n+1} \;=\; X_n + 1_{\{X_n > Y_n\}} B_{n+1} + 1_{\{X_n < Y_n\}} B'_{n+1}, \\ Y_{n+1} \;=\; X_n. \end{cases}$$

4. We can obtain a non-Markovian process starting from a Markov chain of which we only have incomplete knowledge. Let for example $(X_n)_{n\geq 0}$ be

the unbiased random walk on $\mathbb{Z}$ starting at 0, and let $(Z_n)_{n\geq 0}$ be the process which gives the sign of (X_n) as

$$Z_n = \mathbf{1}_{\{X_n>0\}}, \quad n \geq 0.$$

This process is not a Markov chain since, for $n \geq 3$,

$$\mathbb{P}(Z_{n+1} = 1|Z_n = 1, Z_{n-1} = 1, Z_{n-2} = 0) = \\ \mathbb{P}(X_{n+1} > 0|X_n = 2, X_{n-1} = 1, X_{n-2} = 0) = 1,$$

and

$$\mathbb{P}(Z_{n+1} = 1|Z_n = 1, Z_{n-1} = 0, Z_{n-2} = 1) = \\ \mathbb{P}(X_{n+1} > 0|X_n = 1, X_{n-1} = 0, X_{n-2} = 1) = 1/2.$$

Remark 16. *The last example shows that in general, in the definition of Markov chain, we cannot replace the event $X_n = x$ by a less precise event $X_n \in A$, where A is a subset of E. Indeed, it also reads as*

$$\mathbb{P}(X_{n+1} = 1|X_n > 0, X_{n-1} = 1, X_{n-2} = 0) \\ \neq \mathbb{P}(X_{n+1} = 1|X_n > 0, X_{n-1} = 0, X_{n-2} = 1).$$

3.2 Random inductions

Let $(\xi_n)_{n\geq 1}$ be a sequence of i.i.d. random variables taking values in a measurable space I, let X_0 be a random variable taking values in a state space E, independent of ξ_n, $n \geq 1$, and let $f : E \times I \to E$ be a measurable function. We define a process $(X_n)_{n\geq 0}$ with values in E by

$$X_{n+1} = f(X_n, \xi_{n+1}), \quad n \geq 0.$$

The process (X_n) is called a *random induction.*

Proposition 9. *A random induction is a Markov chain, with transition matrix P given for all $x, y \in E$ by*

$$P(x, y) = \mathbb{P}(f(x, \xi) = y).$$

Proof. If $x_0, \ldots, x_{n+1} \in E$ are such that $\mathbb{P}(X_0 = x_0, \ldots, X_n = x_n) > 0$, we have

$$\begin{aligned} &\mathbb{P}(X_{n+1} = x_{n+1}|X_0 = x_0, \ldots, X_n = x_n) \\ &= \mathbb{P}(f(x_n, \xi_{n+1}) = x_{n+1}|X_0 = x_0, \ldots, X_n = x_n) \\ &= \mathbb{P}(f(x_n, \xi_{n+1}) = x_{n+1}). \end{aligned}$$

The last inequality follows from the fact that ξ_{n+1} is independent of $\sigma(X_0, \xi_1, \ldots, \xi_n)$ and $\{X_0 = x_0, \ldots, X_n = x_n\} \in \sigma(X_0, \xi_1, \ldots, \xi_n)$ since X_k is measurable with respect to $\sigma(X_0, \xi_1, \ldots, \xi_k)$ for all $k \geq 0$. Computing $\mathbb{P}(X_{n+1} = x_{n+1}|X_n = x_n)$ gives the same result. This computation also provides the expression for the transition matrix. $\square$

Let now P be a stochastic matrix on E and X_0 be a random variable with values in E. We can construct a Markov process starting from X_0 with transition matrix P as a random induction. Let $(\xi_n)_{n\geq 1}$ be a sequence of i.i.d. random variables uniform on $[0,1]$ and independent of X_0. For all $x \in E$, let also $(I(x,y))_{y\in E}$ be a partition of $[0,1]$ such that $|I(x,y)| = P(x,y)$, and define $f : E \times [0,1] \to E$ by $f(x,u) = y$ where y is the unique element of E such that $u \in I(x,y)$.

Proposition 10. *The random induction defined above is a Markov chain with transition matrix P.*

Proof. By the previous proposition, this random induction is a Markov chain and its transition matrix is given for all $x, y \in E$ by

$$\mathbb{P}(f(x,\xi) = y) \;=\; \mathbb{P}(\xi \in I(x,y)) \;=\; P(x,y).$$

□

An interesting consequence of this result is that, for any stochastic matrix P and random variable X_0 valued in E, we can construct a Markov chain whose transition matrix is P and initial value is X_0.

3.3 Basic properties

Chapman–Kolmogorov equations. The following result shows how to compute the probability of a trajectory of the Markov chain if we know its transition matrix and its initial distribution.

Proposition 11. *Let (X_n) be a Markov chain with transition matrix P and initial distribution μ, i.e., $X_0 \sim \mu$. Then it holds for all $n \geq 0$, and for all $x_0, \ldots, x_n \in E$,*

$$\mathbb{P}(X_0 = x_0, \ldots, X_n = x_n) \;=\; \mu(x_0)P(x_0,x_1)\ldots P(x_{n-1},x_n).$$

Proof. By definition of Markov chain, it follows from conditioning that for any $k \geq 1$,

$$\mathbb{P}(X_0 = x_0, \ldots, X_k = x_k) \;=\; \mathbb{P}(X_0 = x_0, \ldots, X_{k-1} = x_{k-1})P(x_{k-1},x_k),$$

provided that $\mathbb{P}(X_0 = x_0, \ldots, X_{k-1} = x_{k-1}) > 0$. This equality of course remains true when $\mathbb{P}(X_0 = x_0, \ldots, X_{k-1} = x_{k-1}) = 0$. We find the result by applying this relation repeatedly, and observing that $\mathbb{P}(X_0 = x_0) = \mu(x_0)$. □

The Chapman–Kolmogorov equations ensure that the law of a Markov chain is entirely determined by its transition matrix and its initial distribution. In the sequel, we denote by $\mathbb{P}_\mu$ a probability measure such that $(X_n)_{n\geq 0}$ is a Markov chain of transition matrix P and initial distribution μ. If $\mu = \delta_x$, we write simply $\mathbb{P}_x$ instead of $\mathbb{P}_{\delta_x}$. [1]

[1]Given $(X_n)_{n\in\mathbb{N}}$ a Markov chain with transition matrix P and initial distribution ν such that $\nu(x) > 0$ for all $x \in E$, we can define rigorously $\mathbb{P}_x = \mathbb{P}(\cdot|X_0 = x)$ and $\mathbb{P}_\mu = \sum_{x\in E} \mu(x)\mathbb{P}_x$. The fact that $(X_n)_{n\in\mathbb{N}}$ has the desired distribution follows from Chapman–Kolmogorov equation.

Matrix notations. Let E be a state space and let P be a stochastic matrix. Even when E is not finite, we can think of P as a matrix, and define the following operations:

1. If $f : E \to \mathbb{R}$ is a bounded function, we define a new bounded function on E, denoted Pf, by
$$Pf(x) \;=\; \sum_{y\in E} P(x,y)f(y).$$
When E is finite, this corresponds to the matrix multiplication Pf where f is a column vector.

2. If μ is a measure on E, we define a new measure on E, denoted μP, by
$$\mu P(x) \;=\; \sum_{y\in E} \mu(y)P(y,x).$$
When E is finite, this corresponds to the matrix multiplication μP where μ is a row vector.

3. We define by induction a sequence of stochastic matrices $(P^n)_{n\geq 1}$ by
$$P^{n+1}(x,y) \;=\; \sum_{z\in E} P(x,z)P^n(z,y).$$
When E is finite, this corresponds to a matrix multiplication.

Proposition 12. *Let (X_n) be a Markov chain with transition matrix P. Then it holds for all $n \geq 1$,*

1. *for all $x, y \in E$,*
$$\mathbb{P}_x(X_n = y) \;=\; P^n(x,y).$$
2. *for all probability measure μ on E and for all $y \in E$,*
$$\mathbb{P}_\mu(X_n = y) \;=\; \mu P^n(y).$$
3. *for all bounded function $f : E \to \mathbb{R}$ and for all $x \in E$,*
$$\mathbb{E}_x(f(X_n)) \;=\; P^n f(x)$$
where $\mathbb{E}_x$ denotes the expectation under $\mathbb{P}_x$.

Proof. Let us compute
$$\begin{aligned}
&\mathbb{P}_\mu(X_n = y)\\
&= \sum_{x_0,\ldots,x_{n-1}\in E} \mathbb{P}_\mu(X_0 = x_0, X_1 = x_1, \ldots, X_{n-1} = x_{n-1}, X_n = y)\\
&= \sum_{x_0,\ldots,x_{n-1}\in E} \mu(x_0)P(x_0,x_1)\ldots P(x_{n-1},y)\\
&= \sum_{x_0\in E} \mu(x_0)P^n(x_0,y) \;=\; \mu P^n(y).
\end{aligned}$$

Taking $\mu = \delta_x$, we find

$$\mathbb{P}_x(X_n = y) \;=\; \delta_x P^n(y) \;=\; \sum_{z\in E} \delta_x(z) P^n(z,y) \;=\; P^n(x,y).$$

Hence

$$\mathbb{E}_x(f(X_n)) \;=\; \sum_{y\in E} f(y)\mathbb{P}_x(X_n = y) \;=\; \sum_{y\in E} f(y) P^n(x,y) \;=\; P^n f(x).$$

□

Remark 17. *The proof reveals an interesting interpretation of the matrix product: the element $P^n(x,y)$ is equal to the sum of the probabilities of all possible trajectories of the chain which go from x to y.*

Example. Consider a Markov chain on the space $E = \{1,2\}$ with transition matrix P given by

$$P \;=\; \begin{pmatrix} q & p \\ p & q \end{pmatrix},$$

where $0 < p < 1$ and $q = 1-p$. The matrix P is symmetric and can be diagonalized. The eigenvalues are 1 and $1-2p$, and we compute for all $n \geq 1$,

$$P^n \;=\; \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} (1-2p)^n & -(1-2p)^n \\ -(1-2p)^n & (1-2p)^n \end{pmatrix}$$

As $|1-2p| < 1$, we therefore observe that the transition probabilities $P^n(x,y)$ converges exponentially fast to $1/2$ for all states $x, y \in E$.

3.4 Markov properties

Let (X_n) be a Markov chain on the state space E, with transition matrix P.

Markov property. The Markov property expresses the independence between the past and the future given the value of the process at the present time n.

Proposition 13. *Let $n \in \mathbb{N}$, μ a probability measure on E and $x \in E$ such that $\mathbb{P}_\mu(X_n = x) > 0$. Then it holds for all $A \subset E^n$ and $B \subset E^m$ with $m \geq 1$,*

$$\begin{aligned}&\mathbb{P}_\mu((X_0, \dots X_{n-1}) \in A,\, (X_{n+1}, \dots, X_{n+m}) \in B | X_n = x) \\ &\qquad = \mathbb{P}_\mu((X_0, \dots X_{n-1}) \in A | X_n = x)\, \mathbb{P}_x((X_1, \dots, X_m) \in B).\end{aligned}$$

Proof. We develop the left-hand side:

$$\begin{aligned}
&\mathbb{P}_\mu((X_0,\dots X_{n-1}) \in A,\ (X_{n+1},\dots,X_m) \in B | X_n = x)\\
&= \sum_{(x_0,\dots,x_{n-1})\in A} \sum_{(y_1,\dots,y_m)\in B} \frac{\mathbb{P}_\mu(X_0 = x_0,\dots,X_{n-1} = x_{n-1}, X_n = x, X_{n+1} = y_1,\dots,X_{n+m} = y_m)}{\mathbb{P}_\mu(X_n = x)}\\
&= \sum_{(x_0,\dots,x_{n-1})\in A} \sum_{(y_1,\dots,y_m)\in B} \frac{\mu(x_0)P(x_0,x_1)\dots P(x_{n-1},x)P(x,y_1)\dots P(y_{m-1},y_m)}{\mathbb{P}_\mu(X_n = x)}\\
&= \sum_{(x_0,\dots,x_{n-1})\in A} \frac{\mu(x_0)P(x_0,x_1)\dots P(x_{n-1},x)}{\mathbb{P}_\mu(X_n = x)} \sum_{(y_1,\dots,y_m)\in B} P(x,y_1)\dots P(y_{m-1},y_m),
\end{aligned}$$

which is equal to the right-hand side. □

Let us give two consequences of this result:

1. Taking $A = E^n$, we find

   $$\mathbb{P}_\mu((X_{n+1},\dots,X_{n+m}) \in B | X_n = x) \ = \ \mathbb{P}_x((X_1,\dots,X_m) \in B)$$

   In other worlds, conditionally to the event $\{X_n = x\}$, whatever the initial distribution μ, the *shifted* process $(X_{n+m})_{m\geq 0}$ has the same law as the process $(X_m)_{m\geq 0}$ under $\mathbb{P}_x$, *i.e.*, it is a Markov chain with transition matrix P and initial distribution δ_x.

2. Provided that $\mathbb{P}_\mu(X_n = x, (X_0,\dots X_{n-1}) \in A) > 0$, we can write this result in the form

   $$\begin{aligned}
   &\mathbb{P}_\mu((X_{n+1},\dots,X_{n+m}) \in B | X_n = x, (X_0,\dots X_{n-1}) \in A)\\
   &\qquad = \ \mathbb{P}_x((X_1,\dots,X_m) \in B).
   \end{aligned}$$

Strong Markov property. The *strong* Markov property generalizes the previous result in the case where the deterministic time n is replaced by a stopping time T w.r.t. the natural filtration of $(X_n)_{n\in\mathbb{N}}$.

Proposition 14. *Let $x \in E$ and μ a probability measure on E such that $\mathbb{P}_\mu(X_T = x, T < +\infty) > 0$. Then it holds for all $(A_n)_{n\geq 1}$, $A_n \subset E^n$, and $B \subset E^m$ with $m \geq 1$,*

$$\begin{aligned}
&\mathbb{P}_\mu((X_0,\dots,X_{T-1}) \in A_T, (X_{T+1},\dots,X_{T+m}) \in B | X_T = x,\ T < +\infty)\\
&\quad = \ \mathbb{P}_\mu((X_0,\dots,X_{T-1}) \in A_T | X_T = x,\ T < +\infty)\, \mathbb{P}_x((X_1,\dots,X_m) \in B).
\end{aligned}$$

Proof. We use the decomposition $\{T < +\infty\} = \cup_{n\geq 0}\{T = n\}$ to write

$$\begin{aligned}
&\mathbb{P}_\mu((X_0,\dots,X_{T-1}) \in A_T, (X_T,\dots,X_{T+m}) \in B, X_T = x, T < +\infty)\\
&\quad = \ \sum_{n\geq 0} \mathbb{P}_\mu((X_0,\dots,X_{n-1}) \in A_n, (X_{n+1},\dots,X_{n+m}) \in B, X_n = x, T = n).
\end{aligned}$$

For all $n \geq 0$, as $\{T = n\} \in \sigma(X_0, \dots, X_n)$, there exists $\bar{B}_n \subset E^{n+1}$ such that $\{T = n\} = (X_0, \dots, X_n)^{-1}(\bar{B}_n)$ and so

$$\{T = n\} \cap \{X_n = x\} \;=\; \{(X_0, \dots, X_{n-1}) \in B_n, X_n = x\},$$

where

$$B_n = \{(x_0 \dots, x_{n-1}) \in E^n;\; (x_0 \dots, x_{n-1}, x) \in \bar{B}_n\}.$$

Provided that $\mathbb{P}_\mu(X_n = x) \neq 0$, we can apply the Markov property and find

$$\begin{aligned}
&\mathbb{P}_\mu((X_0, \dots, X_{n-1}) \in A_n, (X_{n+1}, \dots, X_{n+m}) \in B, X_n = x, T = n) \\
&= \mathbb{P}_\mu((X_0, \dots, X_{n-1}) \in A_n \cap B_n, X_n = x)\mathbb{P}_x((X_1, \dots, X_m) \in B) \\
&= \mathbb{P}_\mu((X_0, \dots, X_{n-1}) \in A_n, X_n = x, T = n)\mathbb{P}_x((X_1, \dots, X_m) \in B).
\end{aligned}$$

Otherwise this equality still holds as it reads $0 = 0$. Thus we can conclude that

$$\begin{aligned}
&\mathbb{P}_\mu((X_0, \dots, X_{T-1}) \in A_T, (X_T, \dots, X_{T+m}) \in B, X_T = x, T < +\infty) \\
&= \sum_{n \geq 0} \mathbb{P}_\mu((X_0, \dots, X_{n-1}) \in A_n, X_n = x, T = n)\mathbb{P}_x((X_1, \dots, X_m) \in B) \\
&= \mathbb{P}_\mu((X_0, \dots, X_{T-1}) \in A_T, X_T = x, T < +\infty)\mathbb{P}_x((X_1, \dots, X_m) \in B).
\end{aligned}$$

The conclusion follows by dividing this equality by $\mathbb{P}_\mu(X_T = x, T < +\infty)$. $\square$

We can again give two implications of the result:

1. Taking $A_n = E^n$ for all $n \in \mathbb{N}$, we find

   $$\mathbb{P}_\mu((X_{T+1}, \dots, X_{T+m}) \in B | X_T = x,\, T < +\infty) \;=\; \mathbb{P}_x((X_1, \dots, X_m) \in B).$$

   In other worlds, conditionally to the event $\{X_T = x\} \cap \{T < \infty\}$, whatever the initial distribution μ, the *shifted* process $(X_{T+m})_{m \geq 0}$ has the same law as the process $(X_m)_{m \geq 0}$ under $\mathbb{P}_x$, *i.e.*, it is a Markov chain with transition matrix P and initial distribution δ_x.

2. Provided that $\mathbb{P}_\mu(X_T = x,\, T < +\infty, (X_0, \dots, X_{T-1}) \in A_T) > 0$, we can also write this result in the form

   $$\begin{aligned}
   &\mathbb{P}_\mu((X_{T+1}, \dots, X_{T+m}) \in B | X_T = x,\, T < +\infty, (X_0, \dots, X_{T-1}) \in A_T) \\
   &\qquad = \mathbb{P}_x((X_1, \dots, X_m) \in B).
   \end{aligned}$$

3.5 Recurrence and transience

Consider a Markov chain (X_n) on a state space E with transition matrix P. The aim of this section is to classify the states of (X_n) into two categories, namely, transient and recurrent.

Definitions and characterizations. Given $x \in E$, we define the stopping time

$$T_x \;=\; \min\{n \geq 1 : X_n = x\}.$$

It corresponds to the first time of return to x.

Definition. *We say that x is a recurrent state if $\mathbb{P}_x(T_x < +\infty) = 1$ and that x is a transient state if $\mathbb{P}_x(T_x < +\infty) < 1$.*

Let us also set

$$N_x = \sum_{k=1}^{\infty} 1_{\{X_k = x\}}.$$

It corresponds to the total number of return to x. The next result says that the Markov chain returns infinitely often to a recurrent point, and only a finite number of times to a transient point.

Proposition 15. *Let $x \in E$. Then the following characterizations hold:*

1. *The state x is recurrent if and only if $\mathbb{P}_x(N_x = +\infty) = 1$.*
2. *The state x is transient if and only if $\mathbb{P}_x(N_x < +\infty) = 1$. In this case, N_x follows, under $\mathbb{P}_x$, a geometric law with parameter $p = \mathbb{P}_x(T_x = +\infty)$. In particular,*

$$\mathbb{E}_x(N_x) \;=\; \frac{1}{\mathbb{P}_x(T_x = +\infty)} - 1.$$

Proof. We aim to show that

$$\mathbb{P}_x(N_x \geq n) \;=\; \mathbb{P}_x(T_x < +\infty)^n,$$

which will prove the theorem. For $n = 0$, the equality just reads $1 = 1$. For $n \geq 1$, we have

$$\begin{aligned}
\mathbb{P}_x(N_x \geq n+1) \;&=\; \mathbb{P}_x\Big(T_x < +\infty, \sum_{k=1}^{+\infty} \mathbf{1}_{X_{T_x+k}=x} \geq n\Big) \\
&=\; \mathbb{P}_x(T_x < +\infty)\mathbb{P}_x\Big(\sum_{k=1}^{+\infty} \mathbf{1}_{X_k = x} \geq n\Big) \\
&=\; \mathbb{P}_x(T_x < +\infty)\mathbb{P}_x(N_x \geq n),
\end{aligned}$$

where the third equality follows from the strong Markov property. Indeed, it holds

$$\begin{aligned}
\mathbb{P}_x\Big(\sum_{k=1}^{+\infty} \mathbf{1}_{X_{T_x+k}=x} \geq n \Big| T_x < +\infty\Big) &= \lim_{\ell\to\infty} \mathbb{P}_x\Big(\sum_{k=1}^{\ell} \mathbf{1}_{X_{T_x+k}=x} \geq n \Big| T_x < +\infty\Big) \\
&= \lim_{\ell\to\infty} \mathbb{P}_x\Big(\sum_{k=1}^{\ell} \mathbf{1}_{X_k=x} \geq n\Big) \\
&= \mathbb{P}_x\Big(\sum_{k=1}^{+\infty} \mathbf{1}_{X_k=x} \geq n\Big),
\end{aligned}$$

where the second inequality follows from Proposition 14 as $\{T_x < \infty\} = \{T_x < \infty,\, X_{T_x} = x\}$. We conclude immediately by induction. $\square$

We infer from this result a new characterization of recurrence.

Proposition 16. *A state $x \in E$ is recurrent, if and only if,*

$$\mathbb{E}_x[N_x] = \sum_{n\geq 1} P^n(x,x) \;=\; +\infty.$$

Proof. By Proposition 15, a state x is transient if and only if $\mathbb{E}_x[N_x] < +\infty$. In addition, it holds

$$\mathbb{E}_x(N_x) \;=\; \mathbb{E}_x\Big(\sum_{n\geq 1} 1_{\{X_n = x\}}\Big) \;=\; \sum_{n\geq 1} \mathbb{P}_x(X_n = x) \;=\; \sum_{n\geq 1} P^n(x,x).$$

$\square$

Irreducible chains. We say that a state $y \in E$ is *attainable* from a state $x \in E$ if there exists $n \geq 1$ such that $\mathbb{P}_x(X_n = y) > 0$. We say the chain is *irreducible* if all states are attainable from one another,*i.e.*,

$$\forall x, y \in E,\ \exists n \geq 1: \quad \mathbb{P}_x(X_n = y) > 0.$$

Examples.

1. All the examples of Markov chains introduced above (including the random walk on $\mathbb{Z}^d$) are irreducible.
2. The following transition matrices

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P' = \begin{pmatrix} 1 & 0 & 0 \\ 1/3 & 1/3 & 1/3 \\ 0 & 0 & 1 \end{pmatrix}.$$

define chains which are not irreducible on $E = \{1,2,3\}$.

The states of an irreducible chain are all of the same nature.

Proposition 17. *If the chain is irreducible, then either all states are transient, or all states are recurrent.*

Proof. Let $x, y \in E$. Suppose that x is recurrent and let us show that y is also recurrent. We know that there exists $k, j \geq 1$ such that $P^k(x,y) > 0$ and $P^j(y,x) > 0$. For $n > k+j$, we write

$$P^n(y,y) \;\geq\; P^k(y,x) P^{n-k-j}(x,x) P^j(x,y).$$

Setting $\alpha = P^k(y,x)P^j(x,y) > 0$, we find then

$$\sum_{n\geq 1} P^n(y,y) \;\geq\; \sum_{n>k+j} P^n(y,y) \;\geq\; \alpha \sum_{n>k+j} P^{n-k-j}(x,x) \;=\; +\infty.$$

$\square$

We can decide the alternative when E is finite.

Proposition 18. *If the chain is irreducible and the state space E is finite, then all states are recurrent.*

Proof. Let $x \in E$. We have

$$\sum_{y \in E} \sum_{n \geq 1} P^n(x,y) \;=\; \sum_{n \geq 1} \sum_{y \in E} P^n(x,y) \;=\; \sum_{n \geq 1} 1 \;=\; +\infty.$$

As E is finite, there exists $y \in E$ such that $\sum_{n \geq 1} P^n(x,y) = +\infty$. We claim that this implies that y is recurrent. Indeed there exists $k \geq 1$ such that $P^k(y,x) > 0$. Hence $P^n(y,y) \geq P^k(y,x)P^{n-k}(x,y)$ for for all $n > k$ and therefore

$$\sum_{n \geq 1} P^n(y,y) \;\geq\; P^k(y,x) \sum_{n > k} P^{n-k}(x,y) \;=\; +\infty.$$

As all states are of the same type, all states are recurrent. $\square$

Example : the random walk in $\mathbb{Z}^d$. Let $d \geq 1$. The random walk in $\mathbb{Z}^d$ is the Markov chain (X_n) on $E = \mathbb{Z}^d$ of transition matrix P given by

$$P(x,y) \;=\; \frac{1}{2d} 1_{\{x \sim y\}} \quad \forall x, y \in \mathbb{Z}^d$$

where $x \sim y$ means $|x-y|_1 = 1$. This chain is irreducible and we wonder if it is transient or recurrent. One way to know this is to determine whether the sum $\sum_{n \geq 0} P^n(0,0)$ is finite or not.

To compute this sum, let's start by estimating $P^n(0,0)$ using an intuitive (non-rigorous) approach. Let's write

$$X_n \;=\; X_0 + \sum_{k=1}^{n} B_k,$$

where $(B_n)_{n \geq 1}$ is a sequence of i.i.d. uniform variables on $\{\pm e_1, \ldots, \pm e_d\}$ where $(e_1, \ldots, e_d)$ is the canonical basis of $\mathbb{R}^d$. We compute

$$\mathbb{E}_0(|X_n|_2^2) = n\mathbb{E}_0(|B_1|^2) = n,$$

which means that, starting from the origin, X_n typically stays at a distance at most of order $\sqrt{n}$ from the origin. If we imagine that X_n is approximately uniformly distributed in a disk of radius $\sqrt{n}$ around the origin, we find that

$$P^n(0,0) \sim \frac{1}{n^{d/2}} \quad (n \text{ even}),$$

and we deduce that the walk is recurrent in dimension $d = 1$ and $d = 2$ but transient in dimension $d > 2$. This is the result of Pólya.

To make this rigorous, we will compute $P^n(0,0)$ explicitly. We know that $P^n(0,0) = 0$ if n is odd, and we therefore assume from now on that n even. We recall the Fourier decomposition of the indicator at 0 on $\mathbb{Z}^d$:

$$1_{\{0\}}(x) \;=\; \int_{[0,1]^d} \mathrm{e}^{2i\pi x\cdot\theta}\mathrm{d}\theta, \qquad x \in \mathbb{Z}^d$$

where $x \cdot \theta = x_1\theta_1 + \cdots + x_d\theta_d$. We compute

$$\begin{aligned}
P^n(0,0) \;&=\; \mathbb{P}(X_n^0 = 0) \;=\; \mathbb{E}(1_{\{0\}}(X_n^0)) \;=\; \int_{[0,1]^d} \mathbb{E}\left(\mathrm{e}^{2i\pi X_n^0\cdot\theta}\right)\mathrm{d}\theta \\
&=\; \int_{[0,1]^d} \left(\mathbb{E}(\mathrm{e}^{2i\pi\theta\cdot B_1})\right)^n \mathrm{d}\theta \\
&=\; \int_{[0,1]^d} \left(\frac{1}{d}(\cos 2\pi\theta_1 + \cdots + \cos 2\pi\theta_d)\right)^n \mathrm{d}\theta \\
&=\; \int_{[-1/4,3/4]^d} (\varphi(\theta))^n \mathrm{d}\theta
\end{aligned}$$

where we have set

$$\varphi(\theta) \;=\; \left|\frac{1}{d}(\cos 2\pi\theta_1 + \cdots + \cos 2\pi\theta_d)\right|$$

where the introduction of the absolute value is made possible by the fact that n is even.

On the domain $[-1/4, 3/4]^d$, the function φ is only equal to 1 at the points $a = (0, \ldots, 0)$ and $b = (1/2, \ldots, 1/2)$, and is strictly less than à 1 otherwise (it is interesting to consider the domain $[-1/4, 3/4]^d$ rather than $[0.1]^d$). For $\varepsilon > 0$, which we will have to take sufficiently small in the following, we divide the domain of the integral into three disjoint parts:

$$[-1/4, 3/4]^d \;=\; B(a, \varepsilon) \cup B(b, \varepsilon) \cup D$$

where D contains the points of $[-1/4, 3/4]^d$ which are neither in $B(a, \varepsilon)$ nor in $B(b, \varepsilon)$.

On D, there exists $r < 1$ such that $|\varphi(\theta)| \leq r$ for all $\theta \in D$, and therefore

$$\int_D (\varphi(\theta))^n \mathrm{d}\theta \;\leq\; r^n.$$

On $B(a, \varepsilon)$, we use a Taylor expansion of the function $\ln(\varphi)$,

$$\ln(\varphi(\theta)) \;=\; -\frac{2\pi^2}{d}|\theta|_2^2 + \mathcal{O}(|\theta|_2^4), \qquad \theta \to 0,$$

and one computes

$$\begin{aligned}
\int_{B(a,\varepsilon)} (\varphi(\theta))^n \mathrm{d}\theta \;&=\; \int_{B(0,\varepsilon)} \mathrm{e}^{-n\frac{2\pi^2}{d}|\theta|_2^2 + n\mathcal{O}(|\theta|_2^4)}\mathrm{d}\theta \\
&=\; \frac{1}{n^{d/2}} \int_{B(0,\varepsilon\sqrt{n})} \mathrm{e}^{-\frac{2\pi^2}{d}|u|_2^2 + \mathcal{O}(|u|_2^4/n)}\mathrm{d}u.
\end{aligned}$$

For ε small enough, the integral is bounded above and below by strictly positive numbers independent of n. We obtain a similar estimate on $B(b, \varepsilon)$ and we conclude that there exist constants $C_1, C_2 > 0$ such that, for all even n,

$$\frac{C_1}{n^{d/2}} \;\leq\; P^n(0,0) \;\leq\; \frac{C_2}{n^{d/2}}.$$

3.6 Invariant measure and ergodic theorem

3.6.1 Invariant Measure

Let $(X_n)_{n\in\mathbb{N}}$ be a Markov chain on E with transition matrix P. A measure μ on E is said to be *invariant* or *stationary* if

$$\mu P \;= \mu.$$

When μ is a probability measure, it means that, if the chain is distributed according to the law μ at a time n, then it remains so at every subsequent time. In the general case, μ is an invariant measure if

$$\sum_{y\in E} \mu(y)P(y,x) = \sum_{y\in E} \mu(x)P(x,y) \qquad \forall x \in E,$$

and we can interpret this formula as follows: at any point x of E, the mass that arrives at x is equal to the mass that leaves x.

Examples.

1. The counting measure, *i.e.*, $\mu(x) = 1$ for all $x \in \mathbb{Z}^d$, is invariant for the random walk on $\mathbb{Z}^d$.

2. The probability measure $\frac{1}{2}\delta_1 + \frac{1}{2}\delta_2$ is invariant for the transition matrix

   $$P \;=\; \begin{pmatrix} q & p \\ p & q \end{pmatrix},$$

   where $0 < p < 1$ and $q = 1 - p$.

3. The probability measures $\frac{1}{2}\delta_1 + \frac{1}{2}\delta_2$ and δ_3 are invariant for the transition matrix

   $$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

4. Consider the *shift process* on $\mathbb{N}$ defined by

   $$P(x, x+1) = 1, \quad \forall x \in \mathbb{N}.$$

   It has no invariant measure but the trivial null measure, $\mu(x) = 0$ for all $x \in \mathbb{N}$.

Given $x \in E$, recall that

$$T_x \;=\; \min\{n \geq 1 : X_n = x\},$$

and let us define for all $y \in E$,

$$\mu_x(y) \;=\; \mathbb{E}_x\left(\sum_{k=0}^{T_x-1} 1_{\{X_k=y\}}\right) \quad \in \; \mathbb{R}_+ \cup \{+\infty\}.$$

It corresponds to the expected number of passages by y before returning to x.

Lemma 3. *If x is recurrent, then μ_x is an invariant measure, i.e.,*

$$\mu_x P(y) \;=\; \mu_x(y), \quad \forall\, y \in E.$$

Remark 18. If, in addition, $\mu_x(E) = \mathbb{E}_x[T_x] < +\infty$, then $\frac{1}{\mathbb{E}_x[T_x]}\mu_x$ defines an invariant probability.

Proof. Fix $y \in E$. On the one hand, we have by definition,

$$\mu_x(y) \;=\; \mathbb{E}_x\left(\sum_{k\geq 0} 1_{\{X_k=y\}} 1_{\{T_x>k\}}\right) \;=\; \sum_{k\geq 0} \mathbb{P}_x(X_k = y, T_x > k).$$

On the other hand, it holds

$$\mu_x P(y) \;=\; \sum_{z\in E} \mu_x(z) P(z,y) \;=\; \sum_{k\geq 0}\sum_{z\in E} \mathbb{P}_x(X_k = z, T_x > k) P(z,y).$$

For $k \geq 1$, using the Markov property, we find

$$\begin{aligned}
\mathbb{P}_x(X_k = z, T_x > k)P(z,y) \;&=\; \mathbb{P}_x(X_1 \neq x, \ldots, X_k \neq x, X_k = z)P(z,y) \\
&=\; \mathbb{P}_x(X_1 \neq x, \ldots, X_k \neq x, X_k = z, X_{k+1} = y) \\
&=\; \mathbb{P}_x(T_x > k, X_k = z, X_{k+1} = y),
\end{aligned}$$

expression which remains true for $k = 0$. Hence, taking the sum over $z \in E$, we obtain

$$\mu_x P(y) \;=\; \sum_{k\geq 0} \mathbb{P}_x(X_{k+1} = y, T_x > k).$$

Let us now show that $\mu_x(y) = \mu_x P(y)$. If $y = x$, then $\mu_x(x) = 1$ and

$$\mu_x P(x) \;=\; \sum_{k\geq 0} \mathbb{P}_x(T_x = k+1) \;=\; \mathbb{P}_x(T_x < +\infty) \;=\; 1,$$

by recurrence of the state x. If $y \neq x$, then it holds, on the one hand,

$$\mu_x(y) \;=\; \sum_{k\geq 1} \mathbb{P}_x(X_k = y, T_x > k)$$

and, on the other hand,

$$\mu_x P(y) \;=\; \sum_{k\geq 0} \mathbb{P}_x(X_{k+1} = y, T_x > k+1) \;=\; \sum_{k\geq 1} \mathbb{P}_x(X_k = y, T_x > k).$$

$\square$

Proposition 19. *If E is finite, then there exists an invariant probability.*

Proof. Let μ be the law of X_0. Let us define a sequence $(\mu_n)_{n\geq 1}$ of probability measures on E by

$$\mu_n(x) := \mathbb{E}_\mu\left(\frac{1}{n}\sum_{k=1}^{n} 1_{\{X_k=x\}}\right) \;=\; \frac{1}{n}\sum_{k=1}^{n}\mathbb{P}_\mu(X_k=x) \;=\; \frac{1}{n}\sum_{k=1}^{n}\mu P^k(x)$$

for all $x\in E$. We compute that

$$\mu_n P - \mu_n \;=\; \frac{1}{n}(\mu P^{n+1} - \mu P) \xrightarrow[n\to\infty]{} 0.$$

Hence any accumulation point μ_∞ of (μ_n) satisfies $\mu_\infty P = \mu_\infty$. The conclusion then follows by compactness of the set of probability measures on E, which corresponds to

$$\left\{(\mu(x))_{x\in E}\in[0,1]^{\#E};\ \sum_{x\in E}\mu(x)=1\right\}.$$

□

3.6.2 Ergodic Theorem

Before stating the ergodic theorem, let us establish a preliminary result.

Lemma 4. *Let ν be a probability measure on E.*

1. *If $x\in E$ is a transient state, then $\mathbb{P}_\nu(N_x<+\infty)=1$.*
2. *If the Markov chain is irreducible and recurrent then $\mathbb{P}_\nu(T_x<+\infty)=1$ and $\mathbb{P}_\nu(N_x=+\infty)=1$ for all $x\in E$.*

Proof. First, by using the strong Markov property as in Proposition 15, we observe that for all $n\in\mathbb{N}$,

$$\mathbb{P}_\nu(N_x\geq n+1)=\mathbb{P}_\nu(T_x<+\infty)\mathbb{P}_x(N_x\geq n).$$

Letting $n\to\infty$, we deduce that $\mathbb{P}_\nu(N_x=+\infty)=\mathbb{P}_\nu(T_x<+\infty)\mathbb{P}_x(N_x=+\infty)$. In particular, if x is transient, we conclude that $P_\nu(N_x=+\infty)=0$, *i.e.*, $\mathbb{P}_\nu(N_x<+\infty)=1$.

It remains to show that $\mathbb{P}_\nu(T_x<+\infty)=1$ for all $x\in E$ if the chain is irreducible and recurrent. Since

$$\mathbb{P}_\nu(T_x<+\infty) \;=\; \sum_{y\in E}\mathbb{P}_y(T_x<+\infty)\nu(y),$$

it is sufficient to show that $\mathbb{P}_y(T_x<+\infty)=1$ for all $y\in E$. Then we have

$$\begin{aligned}\mathbb{P}_x(N_x<+\infty)&\geq\mathbb{P}_x(T_y<\infty, X_{T_y+k}\neq x\ \forall\, k\geq 1)\\ &=\mathbb{P}_x(T_y<\infty)\mathbb{P}_y(X_k\neq x\ \forall\, k\geq 1)\\ &=\mathbb{P}_x(T_y<\infty)\mathbb{P}_y(T_x=+\infty),\end{aligned}$$

where the second line follows from the strong Markov property. In particular, if the chain is irreducible and recurrent, then $\mathbb{P}_x(T_y < \infty) > 0$ and $\mathbb{P}_x(N_x < \infty) = 0$. Thus $\mathbb{P}_y(T_x = +\infty) = 0$, *i.e.*, $\mathbb{P}_y(T_x < +\infty) = 1$. $\square$

Theorem 8 (Ergodic Theorem)**.** *Let ν be a probability measure on E and let (X_n) be an irreducible Markov chain with initial distribution ν. Then it holds for all $x \in E$,*

$$\frac{1}{n}\sum_{k=1}^{n} 1_{\{X_k = x\}} \xrightarrow[n\to\infty]{} \frac{1}{\mathbb{E}_x(T_x)} \qquad \mathbb{P}_\nu - a.s.,$$

with the convention $1/\infty = 0$.

Proof. Fix the initial distribution ν of the chain. If the chain is transient, as the number of returns to x is almost surely finite under $\mathbb{P}_\nu$ by Lemma 4, the left-hand side tends to 0 and, as $\mathbb{P}_x(T_x = +\infty) > 0$, the right-hand side is 0. Suppose now that the chain is recurrent. Denote $N(n) = \sum_{k=1}^n 1_{\{X_k = x\}}$ the number of return to x before time n. We have to show that

$$\frac{N(n)}{n} \xrightarrow[n\to\infty]{} \frac{1}{\mathbb{E}_x(T_x)} \qquad \mathbb{P}_\nu - \text{a.s.}$$

Let us define the sequence of stopping times $(\tau_n)_{n\geq 0}$ by

$$\begin{aligned} \tau_0 &= 0, \qquad \tau_1 \;=\; T_x, \\ \tau_k &= \min\{n \geq \tau_{k-1} + 1 : X_n = x\}, \quad k \geq 2. \end{aligned}$$

For every $n \geq 1$, we have by definition

$$\tau_{N(n)} \;\leq\; n \;\leq\; \tau_{N(n)+1},$$

and thus

$$\frac{N(n)}{\tau_{N(n)+1}} \;\leq\; \frac{N(n)}{n} \;\leq\; \frac{N(n)}{\tau_{N(n)}}. \tag{1}$$

The conclusion then follows from the law of large numbers. To see it, start by decomposing for $m \geq 2$,

$$\tau_m \;=\; \tau_1 + \sum_{k=2}^{m}(\tau_k - \tau_{k-1}).$$

and observe that the variables $(\tau_k - \tau_{k-1})_{k\geq 2}$ are i.i.d. with the same law as T_x under $\mathbb{P}_x$. Indeed, it holds for all $(k_1, \ldots, k_m) \in \mathbb{N}^m$,

$$\begin{aligned} &\mathbb{P}_\nu(\tau_1 = k_1, \tau_2 - \tau_1 = k_2, \ldots, \tau_m - \tau_{m-1} = k_m) \\ &\qquad = \mathbb{P}_\nu(\tau_1 = k_1, \tau_2 - \tau_1 = k_2, \ldots, \tau_{m-1} - \tau_{m-2} = k_{m-1})\mathbb{P}_x(T_x = k_m) \\ &\qquad = \mathbb{P}_\nu(T_x = k_1)\mathbb{P}_x(T_x = k_2)\cdots\mathbb{P}_x(T_x = k_m) \end{aligned}$$

where the first equality follows from the Markov property and the second from induction. Thus the strong law of large numbers — for non-negative r.v. as the limit might be infinite — implies that

$$\frac{\tau_m}{m} \xrightarrow[m\to\infty]{} \mathbb{E}_x(T_x) \quad \mathbb{P}_\nu - \text{a.s.}$$

In the same way, $\tau_{m+1}/m \to \mathbb{E}_x(T_x)$ $\mathbb{P}_\nu$–a.s. as $m \to \infty$. In addition, according to Lemma 4, $N(n) \to \infty$ $\mathbb{P}_\nu$–a.s. as $n \to \infty$ and so

$$\frac{N(n)}{\tau_{N(n)}} \xrightarrow[n\to\infty]{} \frac{1}{\mathbb{E}_x(T_x)} \qquad \text{and} \qquad \frac{N(n)}{\tau_{N(n)+1}} \xrightarrow[n\to\infty]{} \frac{1}{\mathbb{E}_x(T_x)} \qquad \mathbb{P}_\nu - \text{a.s.},$$

Thus we can conclude by passing to the limit $n \to \infty$ in (1) that

$$\frac{N(n)}{n} \xrightarrow[n\to\infty]{} \frac{1}{\mathbb{E}_x(T_x)} \qquad \mathbb{P}_\nu - \text{a.s.}$$

□

Corollary 2. *If the chain is irreducible, there is at most one invariant probability and it satisfies*

$$\mu(x) = \frac{1}{\mathbb{E}_x(T_x)}, \qquad \forall\, x \in E.$$

Proof. Let μ be an invariant probability. Then the dominated convergence theorem and the ergodic theorem imply

$$\mu(x) = \mathbb{E}_\mu\left(\frac{1}{n}\sum_{k=1}^{n} 1_{\{X_k = x\}}\right) \xrightarrow[n\to\infty]{} \frac{1}{\mathbb{E}_x(T_x)}.$$

□

Remark 19. Consider an irreducible Markov chain admitting an invariant probability measure. Then the ergodic theorem extends to

$$\frac{1}{n}\sum_{k=1}^{n} f(X_k) \xrightarrow[n\to\infty]{} \mathbb{E}_\mu[f(X)] := \sum_{x\in E} f(x)\mu(x), \qquad \mathbb{P}_\nu - \text{a.s.},$$

for any map $f : E \to \mathbb{R}$ such that $\mathbb{E}_\mu[|f(X)|] < +\infty$. This is the basis of the so-called *Monte Carlo Markov Chain* method which provides an alternative to the classical Monte Carlo methods based on the law of large number.

3.6.3 Invariant Probability

Positive and null recurrence. We say that $x \in E$ is *positive recurrent* if

$$\mathbb{E}_x(T_x) \;<\; +\infty.$$

We say that x is *null recurrent* if

$$\mathbb{P}_x(T_x < +\infty) = 1 \qquad \text{and} \qquad \mathbb{E}_x(T_x) = +\infty.$$

Example. All states of the simple random walk on $\mathbb{Z}^d$, for $d = 1$ or $d = 2$, are null recurrent. Indeed, by Pólya's theorem and by translation invariance,

$$\mathbb{E}_x \left(\frac{1}{n} \sum_{k=1}^{n} 1_{\{X_k = x\}} \right) = \frac{1}{n} \sum_{k=1}^{n} P^k(0,0) \xrightarrow[n\to\infty]{} 0$$

for any $x \in \mathbb{Z}^d$. Hence, by the ergodic theorem and the dominated convergence theorem, we find that $1/\mathbb{E}_x(T_x) = 0$.

Theorem 9. *Consider an irreducible Markov chain on a space E. Then the following three assertions are equivalent:*

(i) *there exists a positive recurrent state*

(ii) *all states are positive recurrent*

(iii) *there exists an invariant probability*

In this case, the invariant probability is unique and given by $\mu(x) = \frac{1}{\mathbb{E}_x(T_x)}$.

Proof. The last statement follows immediately from Corollary 2. Then we aim to show that (ii) $\implies$ (i) $\implies$ (iii) $\implies$ (ii), where the first implication is trivial.

(i) $\implies$ (iii) Let $x \in E$ be a positive recurrent state. According to Lemma 3, the measure μ_x is invariant. In addition,

$$\mu_x(E) = \sum_{y \in E} \mu_x(y) = \mathbb{E}_x[T_x] < +\infty.$$

Thus $\mu = \mu_x/\mathbb{E}_x[T_x]$ is an invariant probability.

(iii) $\implies$ (ii) Assume that there exist an invariant probability μ. According to Corollary 2, we have $\mu(x) = 1/\mathbb{E}_x[T_x]$. It remains to check that $\mu(x) > 0$ for all $x \in E$. Since $\sum_{x \in E} \mu(x) > 0$, there exists $x \in E$ such that $\mu(x) > 0$. In addition, for all $y \in E$, there exists $n \in \mathbb{N}$ such that $P^n(x,y) > 0$ by irreducibly. Thus we have

$$\mu(y) = \mu P^n(y) = \sum_{z \in E} \mu(z) P^n(z,y) \geq \mu(x) P^n(x,y) > 0.$$

□

Corollary 3. *If a Markov chain is irreducible on a finite state space, then all states are positive recurrent and the invariant measure is $\mu(x) = 1/\mathbb{E}_x(T_x)$.*

Proof. This is a straightforward consequence of Proposition 19 and Theorem 9. □